WBR perception: understand, record, explain

KNOWN ownership / all gates OPEN

Stereo-inertial perception

Pi 5 / D435i / ROS 2 Jazzy

STM32 retains balance and safety.

Two short lab sessions; analysis at home.

Study revision: 18 October 2026

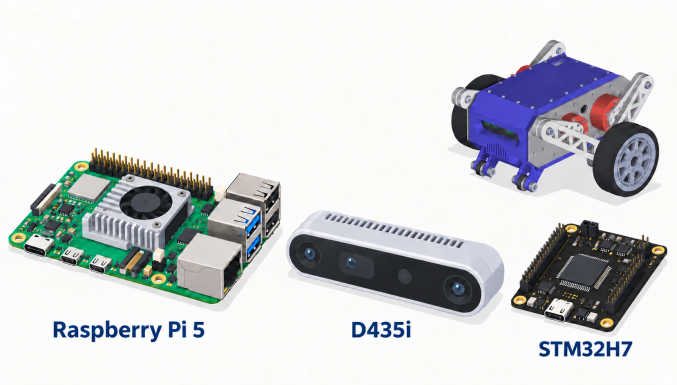


Figure 1: Conceptual platform, not a wiring guide.

> Explain each boundary and leave the lab with evidence, including failures.

How to study this deck

KNOWN method

- Part A: ownership → sensing → runtime → calibration → gates → evaluation.
- Part B: follow step IDs, named terminals, checks and failure branches.
- Recap answers are on the following PDF page; print the session checklists from backup.

> Read the evidence label before believing a number.

What an evidence label means

KNOWN method

Label	Meaning
KNOWN / SOURCE-SUPPORTED	Stable method / inspected source interface.
MEASURED / CALCULATED	Archived observation / arithmetic with stated inputs.
INFERRED / ASSUMED	Interpretation / proposed choice or review threshold.
UNKNOWN / OPEN	Missing evidence / qualification unfinished.

> A layout PASS or script exit 0 cannot promote a hardware gate.

Robot context: what perception contributes

KNOWN method / WBR qualification OPEN

- A wheel-biped robot needs scene-relative motion to navigate beyond its local balance task.
- Perception may support route following and obstacle-aware decisions on the Pi.
- STM32 keeps body sensing, balance, watchdogs and motor limits under local authority.

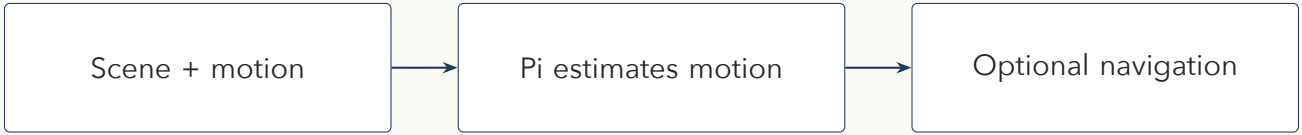


Figure 2: The separate STM32 balance path does not wait for this chain.

Why WBR cares: a navigation aid must have a clear failure boundary.

Research questions become experiments

SOURCE-SUPPORTED project objectives / results OPEN

- Can the Pi sustain the exact stream? Answer with counts, gaps and power logs, before SLAM.
- Does the same frozen take initialize repeatedly? Answer with all run and reset events.
- Is output useful and fresh? Answer with accuracy, continuity, age and receiver rejection tests.

> Write the question, measured quantity and rejection rule before a run.

Sensor-to-pose: follow one image interval

KNOWN method / WBR qualification OPEN

- The SDK acquires sensor frames; the wrapper publishes typed ROS messages.
- A recorder preserves input. A reviewed adapter pairs stereo and attaches covered IMU samples.
- Tracking produces a state; its health and pose epoch must accompany any downstream use.

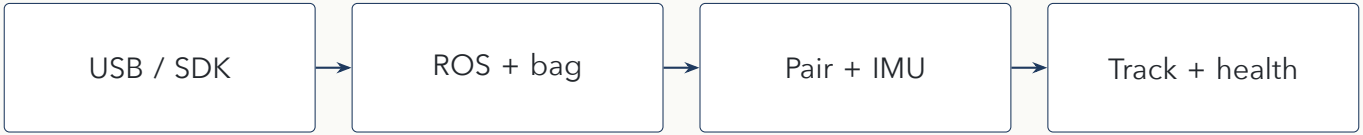


Figure 3: Planned ROS stereo path; current native recording is a separate branch.

Why WBR cares: a fault at an earlier boundary contaminates every later claim.

Machine ownership: Mac, Pi and STM32

KNOWN method / WBR qualification OPEN

- The Mac holds canonical sources and performs home analysis; preserve its existing work.
- The Pi records locally into a separate checksummed runtime and evidence directory.
- SSH carries commands and small evidence. It is not the live image transport for qualification.

> Record source, binary, library, settings and dirty-file hashes for each runtime.

Research design: check yourself

KNOWN method / illustrative question

- A run exits successfully but its pose is 400 ms old. Which research question remains unanswered?

Work out your answer before turning the page.

Research design: answer

KNOWN method / illustrative question

- Freshness remains unanswered. Exit status establishes process completion, not useful source age.
- A WBR receiver needs a bounded pose age and valid session/frame before using optional input.

Explain the evidence that would distinguish the two cases.

System ownership: why WBR needs two computers

KNOWN method

- The Pi runs perception and optional navigation on Linux.
- STM32H7 owns body estimation, balance and motor safety.
- LQR = Linear Quadratic Regulator; VMC = Virtual Model Control.
- The Nucleo-F446RE is a link/supervisor stand-in, not the WBR controller.

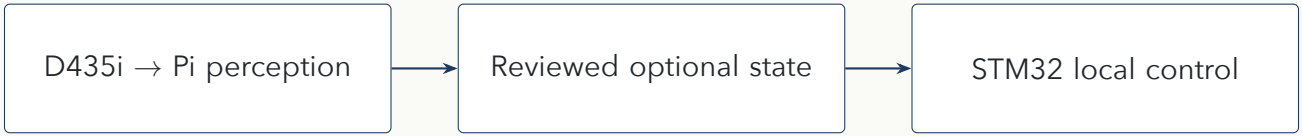


Figure 4: Ownership boundary; no physical safety verdict.

Pi loss must withdraw optional input while local control keeps authority.

Camera IMU and body IMU have different jobs

KNOWN method

- D435i inertial measurement unit (IMU) lives in imu_camera.
- It supports visual-inertial odometry (VIO): motion from images and inertial data.
- The controller IMU lives in imu_body and supports local balance.
- Different mounts, clocks and calibrations prohibit naive substitution.

> An IMU is a sensor type, not a shared reference frame.

Ownership: worked stale-input example

CALCULATED illustrative; no WBR age budget

- Illustrative: 20 ms inference is followed by 300 ms in a queue.
- Even a newly transmitted heartbeat can accompany a 320 ms old result.
- The receiver must assess source age, validity, frame and session.

$$A = 20 + 300 = 320 \text{ ms}$$

Why WBR cares: fast arithmetic does not make a usable navigation input.

Ownership: common mistake and check yourself

KNOWN method

- Mistake: reset the Linux process and assume its old pose is still valid.
- A new process/session may have a new origin, map and clock relation.
- Question: can the camera IMU replace the body IMU during Pi failure?

> Explain the independent control path aloud before connecting anything.

Ownership: answer

KNOWN method / WBR qualification OPEN

- No. The camera path depends on the Pi and has unrelated mount, clock and calibration.
- STM32 keeps independent body sensing and local balance authority.

Use this answer to check the reasoning on the preceding page.

D435i inputs and the identity that matters

MEASURED archived identity / fresh readback OPEN

- New SDK serial 317422076650; firmware readback 5.17.3.10; BMI085.
- Keep SDK, ASIC, firmware-update and USB descriptor serial fields separate.
- Old SDK serial 243322073847 has no transferable gate receipts.

Why WBR cares: a replacement camera needs fresh acquisition evidence.

Two stream profiles: never merge their evidence

SOURCE-SUPPORTED profiles / qualification OPEN

Purpose	IR images	Raw motion
N2 qualification	IR1 Y8 640×480, 15 Hz	Gyro 200; accel 100 Hz
N4 development / S / C	IR1 + IR2 Y8 640×480, 30 Hz	Gyro 200; accel 100 Hz
A Allan / I diagnostics	No images	Gyro 200; accel 100 Hz

N2 still needs 20 → 105 → 300 s, including the 300-second soak.

USB and power: inspect the camera branch

MEASURED historical / SOURCE-SUPPORTED requirement

- D435i goes directly to a blue USB 3 port with its own known cable.
- The camera node must show $\geq$5000M; a 5000M root hub is insufficient.
- The 8 GB Pi booted the old card, but historical under-voltage persisted.

Why WBR cares: missing motion or frames can originate below the estimator.

Power faults propagate into sensor symptoms

KNOWN method / WBR qualification OPEN

- A weak supply can produce resets, USB interruptions or throttling under acquisition load.
- Keep both current fault bits and historical sticky bits; a later clean sample does not erase a fault.
- A newly available 27 W supply still needs loaded evidence on this exact assembly.

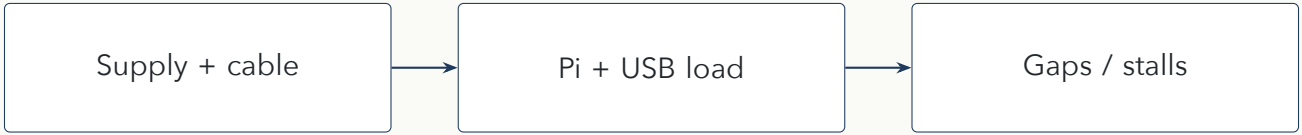


Figure 5: Diagnostic causal possibilities, not a proven cause for every dropped frame.

Check power before changing drivers or estimator settings.

Storage: calculate the payload before recording

CALCULATED payload / MEASURED USB-stick rate

- Each Y8 image has 640×480 bytes; two views arrive 30 times each second.
- The payload alone is 18.432 MB/s before serialization and indexes.
- Two 120 s representations can require over 4.4 GB, before derived files.

$$2(640)(480)(30) = 18{,}432{,}000\ \mathrm{B/s}$$

The 1.09 MB/s USB2 stick cannot record this stream.

Storage qualification: identity and sustained writes

SOURCE-SUPPORTED contract / gate OPEN

Check	What it proves
Mount source + UUID + physical device	The output path reaches the intended data volume.
Overall and worst fsync-batch rate	Cache bursts do not hide long durable-write stalls.
SHA-256 readback + free-space reserve	Data survived the write and future takes have room.

Why WBR cares: a fast sensor can be delayed by a slow recorder.

Storage: worked load and margin

MEASURED historical rate / CALCULATED deficit / ASSUMED margin

- The observed card rate was 16.6 MB/s; stereo image payload is about 18.4 MB/s.
- Illustrative deficit: $(18.4 - 16.6) \times 60 \approx 108$ MB accumulates in one minute.
- The approved card exception proposes $\geq$27.6 MB/s, a 1.5$\times$ margin, plus a take-length test.

$$R_{write} < R_{in} \implies \text{backlog grows until a buffer fills}$$

A higher card capacity does not establish sustained speed or a valid N3 receipt.

Storage review: check yourself

KNOWN method / illustrative question

- Mistake: an empty mount directory exists, so a recording is on the external disk. What must you inspect?

Work out your answer before turning the page.

Storage review: answer

KNOWN method / illustrative question

- Inspect the resolved mount source, partition UUID and physical device, then the qualified write receipt.
- If the mount failed, the same path can resolve to the boot card. Stop before the bag is recorded.

Explain the evidence that would distinguish the two cases.

Hardware: common mistake and check yourself

KNOWN method

- Mistake: a larger RAM board repairs a weak supply or slow card.
- A 15 s RAM diagnostic had zero observed drops; long card writes were gapped.
- Question: does a clean RAM capture qualify persistent storage?

Separate transport, temporary capture, durable copy and replay evidence.

Hardware: answer

KNOWN method / WBR qualification OPEN

- No. A clean RAM capture checks a short acquisition window, not durable storage.
- The archived 15 s RAM test was clean; the longer card take had 0.3–2.9 s gaps after 22 s.

Use this answer to check the reasoning on the preceding page.

Stereo geometry: two views constrain depth

KNOWN method

- Rectification makes corresponding scene points lie on a shared image row.
- Disparity is left u minus right u; baseline is a positive distance.
- Wrong pairing, camera order or focal length makes metric depth wrong.

$$d = u_L - u_R, \qquad Z = \frac{f_x b}{d}$$

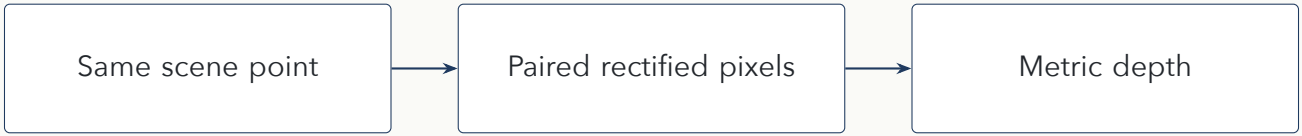


Figure 6: Conceptual stereo reasoning, not a depth measurement.

Why WBR cares: stereo gives scale only with valid geometry.

Features connect stereo views and time

KNOWN method / WBR qualification OPEN

- A feature is a repeatable image location and descriptor, not a sensor measurement of pose.
- Stereo matches constrain depth at one epoch; temporal matches constrain motion between epochs.
- Reject ambiguous correspondences and inspect geometry before increasing feature count.



Figure 7: Stereo and temporal constraints cooperate; neither guarantees a correct match.

Why WBR cares: natural texture must remain observable while the robot moves.

Emitter off: preserve the meaning of a feature

ASSUMED protocol / SOURCE-SUPPORTED rationale

- The projector adds infrared texture tied to the camera rather than fixed scene texture.
- For natural-feature tracking and target calibration, the plan selects emitter OFF.
- Use sufficient ambient near-infrared light. If the target is dark, improve light or print first.

> Log actual emitter and exposure readback; an intended setting is not observed behavior.

Stereo: worked depth and disparity

MEASURED readback / CALCULATED example

- Factory readback: fx = 386.9533 px; baseline = 0.04991255 m.
- Their product is 19.31383 px·m; at 1 m the ideal disparity is 19.31 px.
- At disparity 10 px, calculated depth is about 1.931 m.

$$f_x b = 19.31383\ \mathrm{px\,m}, \quad Z(10\ \mathrm{px}) = 1.93138\ \mathrm{m}$$

The calculation uses readback geometry; it is not independent accuracy.

Stereo: small disparity makes distant depth fragile

CALCULATED first-order illustrative error

- Differentiate Z = fb/d: a small pixel error has increasing depth effect.
- At 2 m, d ≈ 9.657 px; illustrative 0.5 px error gives ≈0.104 m error.
- Lighting, blur, target visibility and stereo skew affect matching.

$$|\delta Z| \approx \frac{Z^2}{f_x b}|\delta d| = \frac{4}{19.31383}(0.5) = 0.104 \text{ m}$$

Use static S intervals before diagnosing a geometric failure.

Stereo: common mistake and check yourself

KNOWN method

- Mistake: copy the SDK negative x translation into Stereo.b.
- SDK IR1→IR2 has tx = −b; ORB baseline magnitude is +b.
- Question: if left/right images are exchanged, what happens to disparity?

Test frame direction and signs before changing feature thresholds.

Stereo: answer

KNOWN method / WBR qualification OPEN

- Exchanging left and right reverses disparity under the same convention.
- The unchanged formula then produces negative depth. Repair ordering and transform direction before tuning features.

Use this answer to check the reasoning on the preceding page.

IMU model: what the two channels measure

KNOWN method

- Gyro measures angular rate plus bias and noise, in rad/s.
- Accelerometer measures specific force, including the gravity relation.
- Bias, scale/misalignment, white noise and bias walk are different errors.

$$\omega_m = \omega + b_g + n_g, \qquad a_m = R_{bw}(a_w - g_w) + b_a + n_a$$

Why WBR cares: systematic IMU errors bend prediction between images.

IMU errors enter prediction before the next image

KNOWN method / WBR qualification OPEN

- Integrating angular rate predicts orientation; rotating specific force helps predict velocity.
- Bias is a systematic offset. White noise and bias walk describe different stochastic behavior.
- An incorrect frame or time interval can look like bad noise tuning.



Figure 8: Conceptual propagation; scale and misalignment are not corrected by this drawing.

Why WBR cares: visual correction has to repair the prediction built from these inputs.

Visual-inertial prediction bridges image intervals

KNOWN method

- At 30 Hz, image intervals are about 33.3 ms; gyro nominal dt is 5 ms.
- Gyro-grid association uses original gyro and bracketed interpolated accel.
- Visual residuals correct accumulated propagation errors.

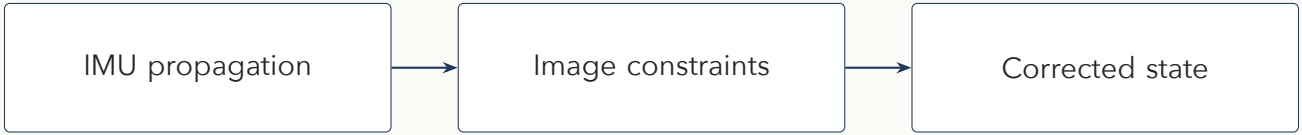


Figure 9: Timing and geometry must connect both sensor cues.

Interpolation adds an aligned input grid, not a new sensor.

IMU: worked bias and noise scaling

CALCULATED illustrative / ASSUMED candidate noise

- Illustrative uncorrected gyro bias 0.01 rad/s accumulates 0.1 rad in 10 s.
- That is about 5.73° before visual correction; it is not this unit's bias.
- Continuous gyro density 0.001 at 200 Hz gives discrete $\sigma \approx 0.01414$ rad/s.

$$\delta\theta = b_g T = 0.1 \text{ rad}, \quad \sigma_d = N\sqrt{f} = 0.001\sqrt{200}$$

ORB scales continuous densities internally; never discretize twice.

Allan deviation: choose the supported region

KNOWN method

- Average stationary samples over τ and compare neighboring averages.
- A −1/2 log slope supports white-noise density N at τ = 1 s.
- A +1/2 slope supports walk K via its fitted value at τ = 3 s.

$$\sigma_A^2(\tau) = \tfrac{1}{2}\,\mathrm{mean}(\bar{x}_{k+m} - \bar{x}_k)^2$$

A flat bias-instability minimum is not the bias-walk coefficient.

Allan plot: read slopes, not just the minimum

CALCULATED schematic / unit noise UNKNOWN

- A log-log slope identifies a candidate noise process over a supported range.
- A short record has few independent long-τ clusters; the last points can mislead.



Figure 10: Illustrative shape; no unit-specific Allan data are plotted.

A fit needs enough clusters and a stable range, not a convenient slope.

Allan: common mistake and check yourself

KNOWN method

- Mistake: use rounded vendor CSV or interpolated 200 Hz accel as raw data.
- Preserve XYZ full precision; raw accel remains 100 Hz and gyro 200 Hz.
- Question: can a clean 36 s pilot establish long-term random walk?

Use the pilot for acquisition; plan $\geq$4 h provisional, preferably 15–24 h.

Allan: answer

KNOWN method / WBR qualification OPEN

- No. A 36 s pilot checks acquisition and some short-time behavior, not supported long-term walk.
- Use $\geq$4 h provisionally, preferably 15–24 h; report unsupported walk as UNKNOWN.

Use this answer to check the reasoning on the preceding page.

Frames: axes and the pose origin

KNOWN method

- Body convention: x forward, y left, z up. Optical: x right, y down, z forward.
- A trajectory describes a named sensor origin in a named reference frame.
- The camera-to-robot mount must be composed to obtain robot-body pose.

Why WBR cares: a good camera trajectory can still give the wrong body motion.

Odometry and mapping answer different questions

KNOWN method / WBR qualification OPEN

- Odometry tracks locally continuous motion, although its accumulated drift may grow.
- Mapping reuses places to improve global consistency; a loop correction can move the map pose.
- Why WBR cares: feeding that jump into a continuous local reference can create an artificial command.



Figure 11: Planned transform semantics, not an implemented WBR continuity bridge.

The map-to-odom continuity bridge needs its own explicit owner.

Robot transforms: assign each edge an owner

KNOWN method / WBR qualification OPEN

- The robot description owns static mounts such as base_link to camera_link and body IMU.
- A future estimator/continuity bridge must explicitly own dynamic map, odom and body edges.
- An external world reference is optional and needs its own documented source.

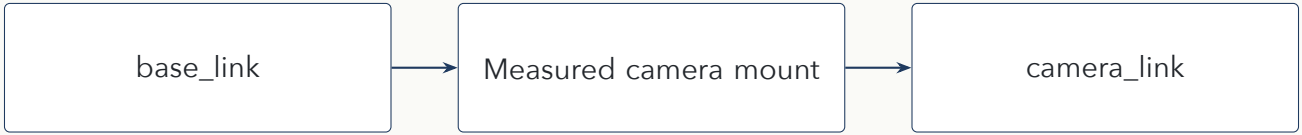


Figure 12: Static robot mount is separate from dynamic pose estimation.

One child frame has one parent and one publisher.

Camera transforms: inspect the wrapper tree

KNOWN method / WBR qualification OPEN

- The RealSense wrapper owns its internal optical and motion-sensor geometry.
- With tf_publish_rate = 0, inspect its static TF and the actual frame IDs.
- The camera-to-body mount remains a separate robot measurement; camera factory data do not supply it.

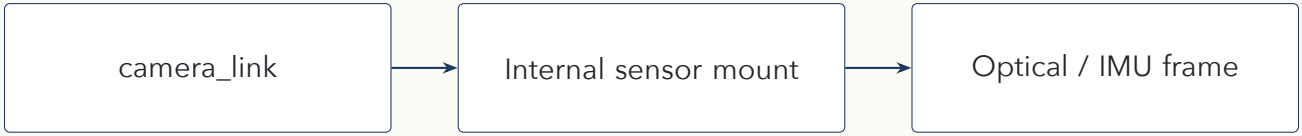


Figure 13: Conceptual wrapper-owned edge; actual names must come from the observed graph.

> Do not publish a copied conceptual sensor tree on top of the wrapper.

Mapping continuity: check yourself

CALCULATED illustrative / bridge OPEN

- Worked case: loop correction moves map pose by 0.20 m while the robot is stationary. Did the robot move?

Work out your answer before turning the page.

Mapping continuity: answer

CALCULATED illustrative / bridge OPEN

- No. The reference estimate changed. A 0.20 m map correction is not evidence of physical body motion.
- The common mistake is to relabel map pose as continuous odom without maintaining the transform between them.

Explain the evidence that would distinguish the two cases.

Transforms: write the direction as an equation

KNOWN method

- T_ab maps coordinates from b into a: p_a = R_ab p_b + t_ab.
- The inverse rotation is transpose; inverse translation is $-R^T t$.
- Keep mathematical rotation direction separate from array storage order.

$$T_{ab}^{-1} = \begin{bmatrix} R_{ab}^T & -R_{ab}^T t_{ab} \\ 0 & 1 \end{bmatrix}$$

A name such as Tbc is useful only with its mapping equation.

Transforms: worked rotated lever arm

CALCULATED illustrative

- Illustrative camera origin is 0.1 m along the body x axis.
- A 90° body yaw rotates that lever arm into the world y axis.
- Copying camera position to base_link creates orientation-dependent error.

$$R_z(90^\circ) \begin{bmatrix} 0.1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.1 \\ 0 \end{bmatrix}$$

Rigid alignment cannot remove a changing lever-arm error.

Frames: common mistake and check yourself

KNOWN method

- Mistake: publish the same TF child from both wrapper and robot description.
- Wrapper owns internal sensor geometry; robot description owns its mount.
- Question: can a loop-corrected map pose be relabeled continuous odom?

Assign one parent and one publisher per child.

Frames: answer

KNOWN method / WBR qualification OPEN

- No. A loop-corrected map pose can jump, whereas odom promises local continuity.
- An explicit owner must maintain map-to-odom and body-mount semantics; that bridge remains OPEN.

Use this answer to check the reasoning on the preceding page.

Timestamp chain: every kind has an owner

KNOWN method

Stamp	Owner / meaning
Physical capture	Actual exposure/measurement event; convention UNKNOWN.
Device / SDK frame	Hardware, global or system domain; keep raw metadata.
Driver / ROS header	Wrapper assigns the message's source epoch.
Publish / subscriber receipt	Different events; instrument on a stated host clock.

A shared unit such as nanoseconds does not imply a shared clock.

Timestamp chain: processing, bag and output

KNOWN method

Stamp	Meaning
Processing start / end	Local queue wait and compute intervals.
ROS bag storage time	Recorder event; never a substitute for header/capture.
Estimator pose epoch	Time the estimated state describes.
Emission / file write / MCU receipt	Availability and receiver events, not pose epochs.

Why WBR cares: measuring compute alone misses old input and queueing.

Clocks: worked same-clock latency

CALCULATED illustrative

- Illustrative host-steady events: receive 10, start 30, end 42, emit 45 ms.
- Queue = 20 ms; compute = 12 ms; emission overhead = 3 ms.
- Receipt-to-output = 35 ms; physical capture age is still UNKNOWN.

$$L_{rx \to o} = (30 - 10) + (42 - 30) + (45 - 42) = 35 \text{ ms}$$

Capture age also needs a measured mapping and uncertainty.

Global time maps a clock; it does not measure exposure

KNOWN method / WBR qualification OPEN

- The SDK global-time option maps device timestamps toward host time.
- Save option readback and each frame domain. A configuration flag does not measure mapping error.
- Arrival minus mapped stamp includes mapping error and sensor/driver delay; it is not pure USB latency.

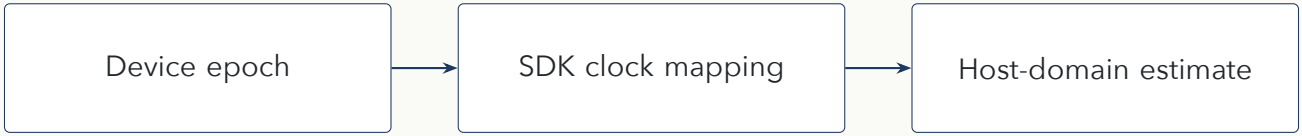


Figure 14: Camera–IMU physical offset is a separate calibration question.

Why WBR cares: usable age needs a clock relation and a stated uncertainty.

Latency decomposition: put boundaries on the drawing

CALCULATED illustrative / capture age UNKNOWN

- Receipt, start, end and output use one host steady clock in this example.
- Queue wait Q, compute C and emission E add to receipt-to-output latency.
- Capture-to-receipt remains outside this interval until the device-to-host mapping is validated.

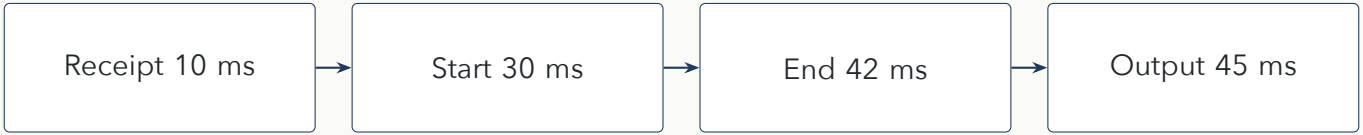


Figure 15: CALCULATED example: Q=20 ms, C=12 ms, E=3 ms; total 35 ms.

A faster compute stage cannot remove a long wait that occurred before it.

Age uncertainty: worked conservative bound

CALCULATED illustrative / WBR budget UNKNOWN

- Illustrative mapped capture time is 100 ms, output is 140 ms, mapping error is ± 3 ms.
- Estimated age is 40 ms; the supported age interval is 37–43 ms.
- If an illustrative receiver limit were 42 ms, the upper bound would not establish freshness.

$$A \in [t_o - \widehat{t}_c - u,\ t_o - \widehat{t}_c + u] = [37, 43]\ \mathrm{ms}$$

Use the conservative upper bound; the actual WBR age limit remains UNKNOWN.

Clocks: common mistake and check yourself

KNOWN method

- Mistake: subtract an MCU timer from a Pi monotonic timestamp.
- Global-time enabled is configuration, not measured physical synchronization.
- Question: all streams say SYSTEM_TIME; is equal-domain checking enough?

Calibration rejects SYSTEM_TIME even when every stream agrees.

Clocks: answer

KNOWN method / WBR qualification OPEN

- No. Equal SYSTEM_TIME domains can still be host-arrival fallback.
- Calibration requires GLOBAL_TIME or HARDWARE_CLOCK plus supported metadata, not just equal domain names.

Use this answer to check the reasoning on the preceding page.

ROS 2: graph, typed topics and QoS

KNOWN method

- Robot Operating System 2 (ROS 2) connects typed publishers and subscribers.
- Data Distribution Service (DDS) provides transport and Quality of Service (QoS).
- QoS does not validate frames or clocks; isolate wrapper faults from the algorithm.



Figure 16: Planned WBR integration; native SDK diagnostics are separate.

Inspect graph, type, QoS and payload together.

QoS: compatibility before counting frames

KNOWN method

Writer	Reader	Match?
Best effort	Reliable	No
Reliable	Best effort	Yes
Volatile	Transient local	No
Reliable + transient local	Same policy for static TF	Yes

Use sensor-compatible QoS and durable static context.

ROS topics: explicit sensor names

SOURCE-SUPPORTED names / observed graph OPEN

Under /camera/camera/	Type / requested rate
infra1/image_rect_raw; infra2/image_rect_raw	Image, 30 Hz each for stereo
infra1/camera_info; infra2/camera_info	CameraInfo, matching image profile
gyro/sample; accel/sample	Imu, raw 200 / 100 Hz
/tf_static (absolute)	TFMessage, reliable / transient local

N3 baseline selects IR1/15 and extra context; IR2 needs separate attestation.

Topic types are part of the input contract

SOURCE-SUPPORTED contract / gate OPEN

ROS 2 type	What to inspect
sensor_msgs/msg/Image	Encoding, width/height, step, frame ID and original header.
sensor_msgs/msg/CameraInfo	Projection/model for that image profile and matching frame.
sensor_msgs/msg/Imu	Raw SI units; gyro and accel are separate sample topics.
tf2_msgs/msg/TFMessage	Static parents/children and durable /tf_static delivery.

> Topic names alone cannot prove compatible contents.

ROS bags: a durable record is an experiment input

KNOWN method

- Choose sqlite3 storage, explicit topics and a fresh take directory.
- Stop with Ctrl-C; wait for metadata finalization before hashing or copying.
- Retain parameter readback, wrapper log, QoS, metadata and power samples.

> Record in the lab; keep interpretation reproducible at home.

ROS: worked queue and common mistake

CALCULATED illustrative / KNOWN ROS semantics

- At 30 Hz, ten queued images can add about 333 ms before processing.
- Mistake: ros2 topic hz looks nominal, so the bag must be complete.
- Question: can a late-starting volatile reader recover prior static TF?

$$10/30 = 0.333\ \mathrm{s}$$

Compare counts, source stamps and context; do not infer them from rate.

ROS: answer

KNOWN method / WBR qualification OPEN

- A nominal ros2 topic hz observation does not prove the recorder stored every sample.
- Reconcile expected, received, stored and replayed counts, original headers, payloads and complete-window gaps.

Use this answer to check the reasoning on the preceding page.

Queues and backpressure: name every buffer

KNOWN method / WBR qualification OPEN

- The SDK, DDS transport, executor, recorder and estimator can each hold data.
- A full queue may block its producer or discard samples; record which policy applies.
- Record arrival/dequeue times, queue depth and discard reason. An unavailable counter is UNKNOWN.

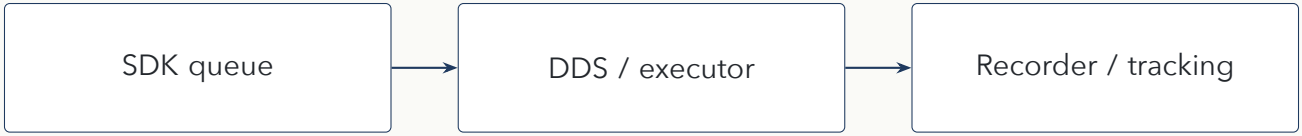


Figure 17: Tracking runs on its caller; mapping and loop workers are separate scheduling loads.

Why WBR cares: intact delivery can still be too old for navigation.

Backpressure: worked overload

CALCULATED illustrative / actual queue policy OPEN

- Illustrative input is 30 images/s and service is 25 images/s: backlog grows by 5 each second.
- After 2 s, 10 queued images need about 400 ms of service before a new image is reached.
- A latest-only policy changes loss and continuity; it is not automatically valid for inertial tracking.

$$\Delta q = (30 - 25) \cdot 2 = 10, \qquad W \approx 10/25 = 0.4\ \mathrm{s}$$

Bound queues and preserve image/IMU coverage; report discarded intervals.

Queue diagnosis: check yourself

KNOWN method / illustrative question

- Mistake: deeper DDS history fixes a recorder that writes too slowly. What happens during sustained overload?

Work out your answer before turning the page.

Queue diagnosis: answer

KNOWN method / illustrative question

- The backlog still grows until the larger buffer fills, and waiting time can become worse.
- First compare durable write rate and input rate; preserve loss and queue evidence instead of changing several buffers.

Explain the evidence that would distinguish the two cases.

Calibration: four questions, four evidence types

KNOWN method

Quantity	Current state
Image intrinsics + stereo geometry	MEASURED factory readback; optical validation OPEN.
Camera–IMU extrinsic	INFERRED nominal CAD; unit alignment OPEN.
Camera–IMU time offset	UNKNOWN; zero is an operating assumption.
IMU densities / walks	ASSUMED example values; unit Allan OPEN.

Why WBR cares: no single calibration report answers all four.

Calibration register: record provenance, not just values

SOURCE-SUPPORTED contract / gate OPEN

Quantity	Store beside the value
Image / stereo geometry	Serial, exact profile/model, readback and optical validation.
Camera-to-IMU / time shift	Transform equation, shift sign, C1–C3 fits and repeatability.
Noise / walk / intrinsic behavior	Raw stationary data, units, fit support and correction state.

A settings file becomes reviewable only when each number has an evidence class.

Calibration: follow data to a candidate

KNOWN method / WBR qualification OPEN

- Measure the target and record the exact profile/options before S and C acquisition.
- Validate timestamps and conversion before fitting geometry or time offset.
- Fit candidates, inspect residuals and compare repeats; validate on held-out data before deployment.

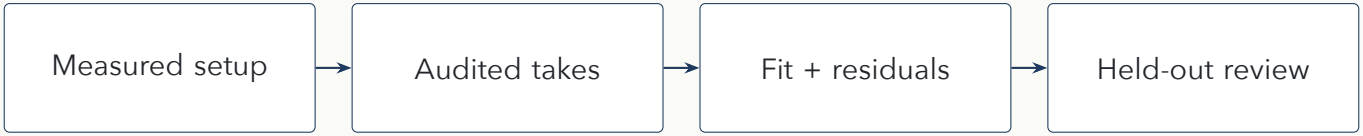


Figure 18: Factory geometry remains operating data; generated candidates are not installed.

Why WBR cares: an optimizer cannot repair wrong units, missing frames or a warped target.

Intrinsics and stereo: validate the operating model

MEASURED readback / ASSUMED operating choice

- Operating candidate is Rectified, 640×480 Y8, factory equal projections.
- Do not refit physical lens distortion from already rectified pixels blindly.
- S take checks rows, fixed-K reprojection, coverage and depth scale.

A fitted PinHole candidate is reviewed separately before deployment.

Measured target: a 6×4 printable AprilGrid

SOURCE-SUPPORTED target / measured dimensions OPEN

- Use A3 45 mm tags, spacing ratio 0.3; A4 30 mm is the fallback.
- Laser print at actual size; measure black edges, axis stretch and flatness.
- Replace nominal tagSize in a session copy; keep tagRows = 4.

> The plan's stale six-row paragraph loses to the actual 6×4 target files.

Camera–IMU calibration: excite observable motion

KNOWN method

- Kalibr uses a fixed measured grid and fixed factory camera chain.
- Record C1–C3 with translation and rotation about all three axes.
- Keep 5–10 s still at each end, emitter off, no impacts, blur or cable pull.

An attractive transform can coexist with bad residuals.

Kalibr transform: worked inversion

SOURCE-SUPPORTED convention / CALCULATED example

- Kalibr T_cam_imu maps IMU→camera; ORB T_b_c1 maps camera→IMU.
- For R = I, inversion negates t; for general R use $-R^T t$.
- ORB b refers to the D435i IMU, never STM32 imu_body.

$$T_{bc1} = T_{cam\,imu}^{-1}, \quad t_{bc1} \approx [-5.52,\ 5.10,\ 11.74]^T \text{ mm}$$

Run basis-vector and inverse-composition checks before using a candidate.

Kalibr time: worked sign and application boundary

SOURCE-SUPPORTED sign / CALCULATED illustrative

- Convention: t_imu = t_cam + shift seconds.
- Illustrative +2 ms maps camera time 1.000 s to IMU time 1.002 s.
- Correct both image stamps once before coverage/association; keep raw audit.

$$t_{cam,corrected} = t_{cam,raw} + 0.002\ \mathrm{s}$$

ORB has no time-offset YAML key; runtime hook availability is OPEN.

Calibration: proposed review thresholds

ASSUMED from current calibration plan

Check	ASSUMED review threshold
S image geometry	RMS $\leq$0.5 px; row median $\leq$0.3 px, p95 $\leq$1 px.
S coverage / depth	$\geq$100 pairs, four quadrants; depth error $\leq$2%.
C repeated extrinsics	Spread $\leq$3 mm / 1°; CAD comparison $\leq$5 mm / 2°.
C time shifts	At least 3 takes; max−min $\leq$1 ms.

These are ASSUMED review thresholds, not manufacturer specs or gate PASS.

Calibration: common mistake and check yourself

KNOWN method

- Mistake: use an optimizer's tiny uncertainty as proof of physical accuracy.
- Check residual patterns, detections, data timing and held-out repeatability.
- Question: can online bias estimates fix unknown IMU scale/misalignment?

> Keep calibration OPEN when the data cannot identify the parameter.

Calibration: answer

KNOWN method / WBR qualification OPEN

- No. Online bias estimates address offsets, not a verified scale or axis-misalignment model.
- Six static gravity poses diagnose accelerometer behavior; gyro scale needs known-rate or observable dynamic evidence.

Use this answer to check the reasoning on the preceding page.

ORB-SLAM3: work shared across threads

KNOWN method

- SLAM = simultaneous localization and mapping; ORB = Oriented FAST and Rotated BRIEF.
- Tracking handles each stereo input, IMU prediction and pose optimization.
- Local mapping optimizes nearby keyframes and scene points.
- Loop closing corrects revisits; Atlas may hold multiple maps.

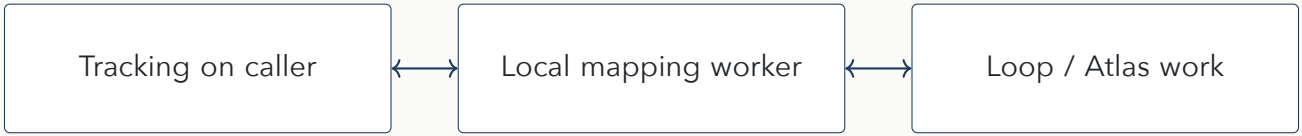


Figure 19: Cooperating roles; bidirectional links show shared work.

Why WBR cares: a nonempty trajectory may hide resets or a map switch.

Tracking: account for every image interval

KNOWN method

- Pair left/right by original identity and valid time relation.
- Supply covered IMU samples; reject duplicate/backward time and gaps.
- Log tracking state, lost frames, resets, map identity and initialization.

> A viewer or successful launch is not a tracking verdict.

Initialization: stereo scale does not solve everything

KNOWN method

- Stereo baseline supplies metric geometry under valid calibration.
- Gravity, velocity and inertial biases still need sufficient excitation.
- A still take can end cleanly with no initialized map.

Use smooth multi-axis motion; keep target calibration and VIO routes separate.

Initialization: check yourself

KNOWN method / illustrative question

- A 15 s still take exits cleanly with no map. Does this establish successful inertial initialization?

Work out your answer before turning the page.

Initialization: answer

KNOWN method / illustrative question

- No. A clean exit and an initialized map are different observations.
- Use smooth excitation, valid timing and initialization events. Still data can test acquisition without identifying all inertial state.

Explain the evidence that would distinguish the two cases.

Association: worked interpolation

CALCULATED illustrative

- Illustrative accel values: 1.0 at 10 ms and 3.0 at 20 ms.
- At gyro time 15 ms, bracketed linear interpolation gives 2.0.
- Outside the bracket or across an invalid gap, reject rather than extrapolate.

$$a(15) = 1 + \frac{15 - 10}{20 - 10}(3 - 1) = 2$$

Preserve both raw accel and the derived gyro-grid value.

Mapping and Atlas: distinguish map state from file output

KNOWN method / WBR qualification OPEN

- Keyframes retain selected views, poses and constraints; local mapping improves nearby structure.
- Atlas can start another map after loss and later merge maps or accept loop corrections.
- A nonempty trajectory can hide multiple maps, resets or relocalizations.

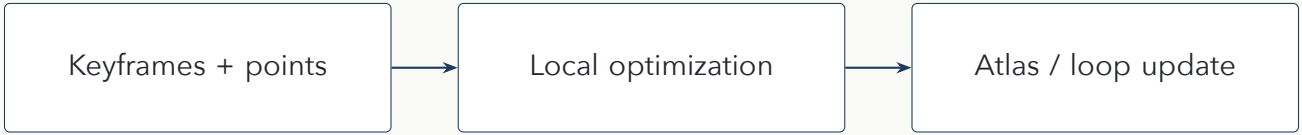


Figure 20: Reference changes require state and map IDs; they are not just visual improvements.

Why WBR cares: downstream state must be re-established after reference changes.

Atlas: worked reset example

CALCULATED illustrative / actual event instrumentation OPEN

- Illustrative run creates map 7, loses tracking, then initializes map 8 at a new origin.
- Both exported segments may contain finite poses; joining their xyz values makes a fictitious path.
- Keep map/session/reset IDs and review the relation before comparing or forwarding segments.

> Common mistake: a continuous file is assumed to be a continuous coordinate frame.

Atlas state: check yourself

KNOWN method / illustrative question

- A trajectory has more rows after relocalization. Does that prove one continuous successful trial?

Work out your answer before turning the page.

Atlas state: answer

KNOWN method / illustrative question

- No. Row count alone cannot reveal map changes, resets or the duration of lost tracking.
- Retain state events and all attempts, then evaluate each supported frame segment.

Explain the evidence that would distinguish the two cases.

Settings audit before feature tuning

SOURCE-SUPPORTED contract / gate OPEN

Parser group	Verify first
Camera / Stereo	Model, selected rectification, dimensions, focal pixels and positive baseline.
IMU.T_b_c1	Camera→estimator-IMU direction; metres; matrix layout and basis tests.
IMU frequency / noise	Input grid and continuous units; no double discretization.
ORBextractor	Only then inspect feature count, pyramid and FAST thresholds.

A settings change must answer a measured failure, using the pinned parser.

Numeric IMU settings: examples are not measurements

ASSUMED candidate values / SOURCE-SUPPORTED units

Key	ASSUMED example / continuous-time units
IMU.NoiseGyro	0.001 rad/s/$\sqrt{\text{Hz}}$
IMU.NoiseAcc	0.01 m/s^2/$\sqrt{\text{Hz}}$
IMU.GyroWalk	0.000001 rad/s^2/$\sqrt{\text{Hz}}$
IMU.AccWalk	0.0001 m/s^3/$\sqrt{\text{Hz}}$

Unit-specific values remain UNKNOWN until supported Allan fitting.

Numeric IMU contract: scaling happens exactly once

SOURCE-SUPPORTED scaling / CALCULATED example

- IMU.Frequency = 200 describes the reviewed combined grid, not the raw 100 Hz accelerometer.
- The pinned tracker multiplies noise density by $\sqrt{f}$ and divides walk by $\sqrt{f}$.
- At 200 Hz, example GyroWalk 1e-6 becomes about 7.07e-8 per step before covariance construction.

$$\sigma_{n,d} = N\sqrt{f}, \qquad \sigma_{w,d} = K/\sqrt{f}$$

Keep the YAML continuous; interpolation correlation is still a model limitation.

ORB: common mistake and check yourself

KNOWN method

- Mistake: increase feature count to hide dropped data or wrong geometry.
- Debug power → USB → raw streams → clocks → ROS/bag → calibration → ORB.
- Question: does offline replay establish live throughput or fresh output?

Use immutable inputs to isolate code changes, then test live scheduling.

ORB: answer

KNOWN method / WBR qualification OPEN

- No. Offline replay does not establish live deadlines, throughput or fresh output.
- Acquire loaded live queue/age/drop and platform traces after immutable offline input is validated.

Use this answer to check the reasoning on the preceding page.

Gate ladder: prerequisites remain OPEN

SOURCE-SUPPORTED current project state

Gate	Meaning / current evidence
N0 / N1	Stable platform / current-unit access; OPEN.
N2 / N3	Exact raw soak / ROS record-replay; OPEN, active N3.
N4A	Known EuRoC dataset on Pi; strong historical evidence, OPEN.
N4B / N4C	D435i offline / live; no qualified camera trajectory.

N4A is a separate branch; it cannot accept the camera prerequisites.

Gate ladder: measurement and controller boundary

OPEN / N5–N7 not started

Gate	Required evidence
N5	Reproducible accuracy/continuity/latency/resources.
N6	Health semantics: stale, invalid, reset, unknown-age fixtures.
N7	Staged controller integration, independent local control.

WBR age budgets and physical safety bounds remain UNKNOWN.

Gate dependencies: branch on the experiment input

KNOWN method / WBR qualification OPEN

- N0 supports the independent EuRoC N4A baseline; that dataset does not test D435i acquisition.
- Camera N4B requires N1–N3 plus stereo extension and calibration; N4C adds live scheduling.
- N5 measurement, N6 semantics and N7 controlled integration need separate evidence.

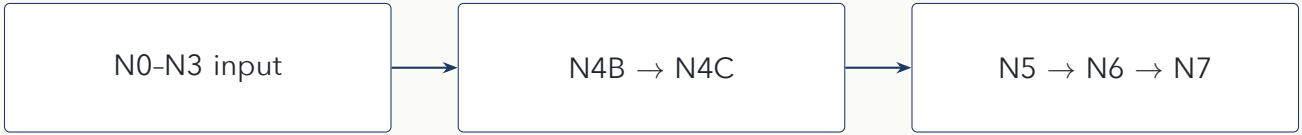


Figure 21: N4A is a separate N0→EuRoC branch; every current gate remains OPEN.

Why WBR cares: identify the claim before deciding which prerequisites apply.

N0 acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Required evidence	If it fails
ARM64, Ubuntu 24.04, ROS 2 Jazzy	Reconcile actual OS/package inventory before sensor work.
UTC synchronized; power flags 0x0	Inspect time service or supply/cable; retain sticky faults.
Temperature <75°C; available RAM ≥512 MiB	Inspect cooling and competing memory load.

Idle readiness must be rechecked under capture and estimator load.

N1 acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Required evidence	If it fails
Camera own direct branch ≥5000M	At 480M inspect cable, port and hubs before streaming.
Video and HID/IIO access; no competing owner	Inspect exact device permissions and process ownership.
Sealed identity/host/boot receipt ≤2 h old	Renew after reboot/topology change; separate serial fields.

A root hub at 5000M does not qualify its camera child.

N2 content, health and actual acceptance

SOURCE-SUPPORTED contract / gate OPEN

Required evidence	If it fails
20 / 105 / mandatory 300 s, clean shutdown	Keep the failed stage and stop downstream acceptance.
Coherent Y8 images; finite plausible motion	Inspect pixels, dimensions, units and exact stream profile.
Clean kernel/SDK and 1 Hz platform health	Return to power/USB before blaming an estimator.
Human review; warning and startup offsets retained	Leave OPEN until the actual review is complete.

N2 rate bands on the next page are necessary but not sufficient.

N3 acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Required evidence	If it fails
60 s live plus delivery under recorder load	Power/raw first, then graph/QoS and durable storage.
Exact ten-topic baseline; parameters/types/TF	Inspect actual graph and durable context, not launch intent.
Final bag and exact complete replay	Reconcile count, order, header and decoded payload hashes.
No live contamination; clean player/observer stop	Stop the camera and isolate replay in domain 43.

The strict baseline is IR1/15; stereo/30 needs separate evidence.

N3 content checks beyond delivery rates

SOURCE-SUPPORTED runbook proposals / no gate PASS

Runbook proposal	Interpretation / failure branch
Stationary mean gyro norm $\leq$0.10 rad/s	Use finite raw vectors and a declared stationary interval.
Stationary mean accel norm 9.0–10.5 m/s^2	Check raw units and frame before claiming plausible gravity.
Zero incomplete/corrupt notifications	Inspect the unfiltered startup-to-stop SDK/kernel logs.

These engineering proposals do not establish calibrated camera–IMU timing.

Storage receipts: old baseline and newer exception

SOURCE-SUPPORTED old contract / ASSUMED newer margin / OPEN

Policy	Requirement / unresolved boundary
Archived strict N3	External UUID; $\geq$1 GiB and $\geq$60 s; overall and worst 64 MiB batch $\geq$9.216 MB/s.
Archived readback	Matching SHA-256, cleanup and 10 GiB retained headroom.
Approved 128 GB-card exception	Take-length/larger-file test $\geq$27.6 MB/s; $\geq 2\times$ take plus OS reserve.

The old runner needs reviewed card-receipt compatibility; never manufacture acceptance.

Stereo extension: preserve the accepted baseline

KNOWN method / WBR qualification OPEN

- N2/N3 qualify IR1 at 15 Hz with raw motion; keep those records unchanged.
- The stereo extension adds IR2, 30 Hz paired load, both projections and pairing evidence.
- It must recheck bandwidth, storage, stamps and geometry rather than inherit an IR1 verdict.

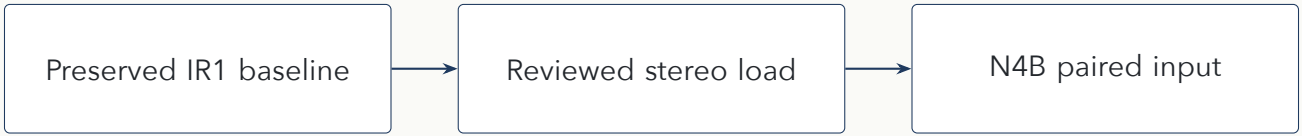


Figure 22: One branch extends the input contract; it does not rewrite old evidence.

A mono-inertial adapter or an IR2 topic name does not establish stereo readiness.

N4A: known-dataset acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Contract	Evidence / failure branch
N0; pinned binary, libraries, settings and MH01	Build/link error → dependencies; empty images → reader paths.
Finite ordered exports and supported coverage	Audit dataset epochs, origin, quaternion and input hashes.
Repeated initialization / all attempts	No inertial init → settings, timing and motion.
ATE/RPE and continuity budgets UNKNOWN	Measured ATE supports the build; power and review keep gate OPEN.

N4A never qualifies camera transport or unit calibration.

N4B: offline D435i acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Contract	Evidence / failure branch
N3 + stereo extension + calibrated inputs	Direct SDK diagnostics cannot bypass formal prerequisites.
Lossless pairs, valid time and IMU coverage	Vertical mismatch → geometry; missing brackets → raw motion.
Geometry, noise and time offset supported	Review residuals/held-out repeats before feature tuning.
Repeatable finite state; budgets UNKNOWN	Retain crashes and resets; absent reference means accuracy UNKNOWN.

A build, short recording or still replay is not a qualified motion trajectory.

N4B provenance: connect raw data to each pose

KNOWN method / WBR qualification OPEN

- The native recorder creates stereo_imu.db3 and timestamps.csv; ordinary ROS bags use a different schema.
- The conversion manifest links raw hash, converter version, pixels, exact ns and settings.
- Retain rejected samples and interpolation decisions; do not sort or invent missing motion.

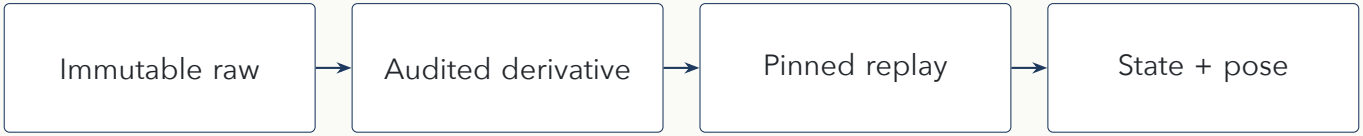


Figure 23: Native and ROS paths need separately validated adapters.

Why WBR cares: replay comparisons are useful only when the input has stayed fixed.

N4C: live acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Contract	Evidence / failure branch
Accepted N4B; renewed stereo acquisition	Same geometry/settings, fresh physical receipts under load.
Initialization and continuity budgets UNKNOWN	Retain eligible attempts, lost intervals and resets.
Latency and queue bounds UNKNOWN	Same-clock events; mapped capture age includes uncertainty.
Offline works, live lags	Inspect wrapper/DDS/executor/queues, then loaded power/thermal.

Live camera mode remains unattempted in the controlling Phase-8 evidence.

N5: measurement acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Contract	Evidence / failure branch
Valid reference for ATE/RPE	Audit frames, association, overlap and rigid alignment.
P50/P95/P99, init/loss, sample counts	Name event interval and clock; averages hide tail stalls.
CPU/thermal/RAM, losses and queue series	Join events; unavailable counters remain UNKNOWN.
Frozen hypotheses and all attempts	At least three offline repeats is a planned minimum.

Numeric performance budgets remain UNKNOWN; a report is not automatic acceptance.

N6: health acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Contract	Evidence / failure branch
Reject every declared invalid fixture	Expected/actual state and reason, on synthetic event time.
Source-age bound $\leq$Amax; Amax UNKNOWN	Clock relation, wrap and delay fixtures; receipt alone is insufficient.
Explicit reset/session/frame re-establishment	Old session accepted $\rightarrow$ protocol; wrap failure $\rightarrow$ receiver clock.

A reviewed receiver harness is planned; the CN demo is not WBR N6 acceptance.

N7: boundary acceptance and diagnosis

SOURCE-SUPPORTED contract / gate OPEN

Contract	Evidence / failure branch
Perception has no safety authority	Ownership/code review plus controlled fault trace.
Local control retained when Pi is lost	STM32 loop/watchdog/motor-state trace on MCU clock.
Fallback inside approved budget; bound UNKNOWN	Measured fault-to-local-action interval and safety review.

Receiver bench → approved controlled integration → separately approved physical test.

N2 thresholds stay exactly as specified

SOURCE-SUPPORTED gate criteria

Stream / condition	Requirement
IR1 / gyro / accel	14.25–15.75 / 190–210 / 95–105 Hz.
Image / motion max gap	$\leq$0.20 s / $\leq$0.05 s.
Order and image identity	Positive time/frame steps; no missing image frames.
Durations and health	20 $\rightarrow$ 105 $\rightarrow$ 300 s; clean power/kernel; actual review.

A 20 s stereo diagnostic is not the 20 s N2 stage.

Dated evidence: what the sources actually establish

MEASURED source-reported diagnostics / all gates OPEN

- Phase 8: card gaps, short RAM success, first-frame crash, interrupted repair.
- Later lab source reports rebuild + 15 s still EuRoC-route replay exit 0.
- That still replay has no map; hand-motion tracking and live remain unverified.

> Keep the later report separate from the Phase-8 historical snapshot.

Gate example, common mistake and check yourself

KNOWN method

- Worked case: 15 Hz interior rate with a long start delay misses window coverage.
- Compare N/T and (N−1)/(last−first), plus start/tail silence.
- Question: may a historical gate receipt survive a camera swap or reboot?

> No tuning can repair evidence from the wrong device or boot.

Gate receipts: answer

KNOWN method / WBR qualification OPEN

- No. A camera swap invalidates unit evidence; reboot or topology change requires a fresh N1.
- N1 is identity/host/boot bound and $\leq$2 h old. The full-request rate and boundaries still need inspection.

Use this answer to check the reasoning on the preceding page.

Worked window check: nominal interior, missing coverage

CALCULATED illustrative / SOURCE-SUPPORTED N2 limit

- Illustrative request lasts 20 s, but 150 images arrive only from t=10 to t=19.933 s.
- Interior rate is about 15 Hz. Whole-window rate is 150/20 = 7.5 Hz.
- Initial silence is 10 s, far beyond the N2 image-gap requirement of 0.20 s.

$$r_{whole} = N/T, \quad r_{interior} = (N-1)/(a_N - a_1), \quad B = \max(a_1, T - a_N)$$

A nominal burst rate cannot accept the requested acquisition window.

Evaluation: the measured EuRoC MH01 baseline

MEASURED historical / N4A OPEN

- Historical 4 GB Pi completed three runs with IMU initialization, one map, no resets.
- Absolute Trajectory Error (ATE), rigid-aligned: 0.0415 / 0.0368 / 0.0416 m.
- These are root mean square errors (RMSE); mean RMSE is about 0.040 m.
- Alignment fits rotation and translation only; scale is not fitted.

Why WBR cares: known input tests the build without camera adaptation.

ATE: worked RMS after a declared alignment

CALCULATED illustrative

- Illustrative position residuals after SE(3) alignment are 0.03 and 0.04 m.
- RMSE is sqrt((0.03^2 + 0.04^2)/2) = 0.03536 m.
- Report association tolerance, overlap, excluded frames and failed trials.

$$\mathrm{ATE}_{RMSE} = \sqrt{\frac{1}{n}\sum_i \|p_i^{ref} - (Rp_i^{est} + t)\|^2}$$

An alignment removes gauge freedom; it does not prove frame compatibility.

RPE: relative pose error over a fixed interval

CALCULATED illustrative / historical RPE UNVERIFIED

- Illustrative reference relative translation is 1.00 m; estimate is 1.02 m.
- With equal rotation/direction, translation RPE is 0.02 m for that pair.
- Declare delta seconds/metres/frames, endpoint policy and eligible pair count.

$$E_{ij} = (T_i^{-1} T_j)_{ref}^{-1} (T_i^{-1} T_j)_{est}$$

Numeric historical RPE remains UNVERIFIED under Phase-8 source restrictions.

Without ground truth: useful proxies have limits

ASSUMED review protocol / accuracy UNKNOWN

- Record initialization time, resets/maps, loss fraction and longest lost interval.
- Use a measured straight segment and return-to-start route with placement uncertainty.
- Loop closure can conceal drift: report pre-loop and final closure separately.

> Tape scale and closure are proxies, never D435i ATE.

Evaluation: common mistake and check yourself

KNOWN method

- Mistake: fit Sim(3) scale and call the metric-scale stereo result accurate.
- A plausible path or low aligned residual can hide scale, frame and timing errors.
- Question: can 184 s dataset time / 282 s wall time give per-frame P99?

Report metric boundaries and evidence eligibility before comparing numbers.

Evaluation: answer

KNOWN method / WBR qualification OPEN

- No. Dataset time divided by wall duration cannot recover per-frame P99 latency.
- Wall duration mixes vocabulary loading, tracking and shutdown; keep raw event boundaries on a named clock.

Use this answer to check the reasoning on the preceding page.

Trajectory conversion: audit one row first

KNOWN method / WBR qualification OPEN

- TUM rows contain seconds, xyz in metres, then quaternion qx qy qz qw.
- EuRoC reference uses ns and qw-first fields. The ORB output format depends on the selected example.
- Check pose direction and physical origin before converting; invert a world-to-camera pose when required.



Figure 24: Preserve originals and save converter version, argv and rejected rows.

Why WBR cares: a converter can create a smooth but physically wrong trajectory.

Trajectory conversion: worked units and quaternion

CALCULATED illustrative / format contract

- Illustrative input epoch 1,000,000,000 ns becomes 1.000000000 s.
- Identity quaternion qw,qx,qy,qz = 1,0,0,0 becomes TUM qx,qy,qz,qw = 0,0,0,1.
- Mistake: leave ns in a seconds field or reorder quaternion values without changing their meaning.

Save an auditable sample row and confirm rotation normalization.

evo: define the metric before invoking it

SOURCE-SUPPORTED source example / execution OPEN

- The source example uses evo_ape with rigid –align for metric stereo trajectories.
- Its evo_rpe example selects angle_deg and a 1 m interval; this is an example, not an acceptance limit.
- Check installed help/version and declare tolerance, pair policy, reference coverage and saved outputs.

H7 supplies the audited first-party evaluator; its RPE pairing is not automatically evo-equivalent.

Percentiles: describe the tail as well as the middle

KNOWN method / WBR qualification OPEN

- P50 is the median, P95 covers 95% and P99 covers 99% under the declared estimator.
- Name the interval: queue wait, compute, receipt-to-output or supported capture age.
- State sample count, warm-up exclusions, failed runs and the percentile method.

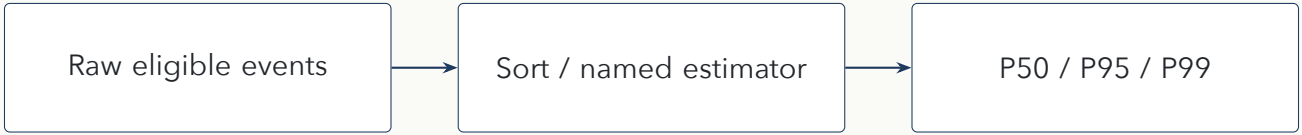


Figure 25: Do not turn dataset-duration divided by wall-time into a latency percentile.

Why WBR cares: occasional old output can matter even when median compute is fast.

Percentiles: worked nearest-rank example

CALCULATED illustrative / nearest-rank estimator

- For 100 ordered eligible latencies, nearest-rank P95 is sample 95 and P99 is sample 99.
- If samples 1–94 are 10 ms and 95–100 are 200 ms, P50=10 ms but P95=P99=200 ms.
- With very few samples, a tail percentile has weak support; retain the full event series.

$$\mathrm{rank}(P_p) = \lceil (p/100)N \rceil$$

Common mistake: report the mean and hide the six slow events.

Loss and continuity: declare the denominator

CALCULATED illustrative / actual continuity OPEN

- Reconcile generated, received, recorded and processed IDs at each boundary.
- Illustrative 3 missing out of 300 expected inputs is 1% loss; 297/10 s is not a zero-loss proof.
- Count every initialization attempt and lost interval; keep resets, map IDs and trajectory coverage.

> An absent drop counter remains UNKNOWN even when observed output is regular.

N5 resource series: join causes to events

SOURCE-SUPPORTED contract / gate OPEN

Series	Interpretation
CPU utilization + clock frequency	State the core normalization; clock throttling changes capacity.
SoC temperature + raw throttling flags	Join warm-up and faults to queue/compute stalls.
RSS + MemAvailable + free data space	Separate process footprint, system pressure and disk headroom.
Queue depth + discard/loss events	Keep clock, units and missing-counter labels explicit.

One-Hz health helps context; it can miss a short event and is not an IMU-temperature sensor.

Historical N4A timing and resource evidence

MEASURED historical / performance qualification OPEN

Observed summary	Practical limit
301 / 282 / 284 s wall for 184 s dataset	Includes load/start/stop; no per-frame percentile.
Peak RSS 823 / 828 / 828 MiB	A peak is not a memory-pressure time series.
87 / 178 live power-fault samples; max 63°C	Power faults invalidate a processor-speed verdict.

Preserve measured summaries while leaving unmeasured timing and joined series UNKNOWN.

Measurement design: check yourself

CALCULATED illustrative

- Worked case: 20 ms compute follows 300 ms queue wait. May you report 20 ms as end-to-end latency?

Work out your answer before turning the page.

Measurement design: answer

CALCULATED illustrative

- No. Receipt-to-output is at least 320 ms before emission overhead; physical capture age needs a clock mapping.
- The mistake is to choose the fastest boundary after looking at the result.

Explain the evidence that would distinguish the two cases.

Pi↔STM32 link: integrity before semantics

KNOWN method

- Step 9b bench firmware checks RTT, CRC, sequence, session and silence.
- It runs on F446RE as a link stand-in; NO_LINK is not OS HOLD or FAILSAFE.
- Why WBR cares: packet integrity and fallback policy are distinct tests.

> Never flash link-bench firmware and expect the CN/OS supervisor.

Planned WBR health envelope

SOURCE-SUPPORTED design intent / WBR interface OPEN

Field family	Receiver purpose
Version, session and sequence	Reject incompatible data, duplicates and old boots.
Source epoch/domain and age bound	Separate new transport activity from a fresh estimate.
Frames, map/reset ID and tracking state	Reject a wrong reference or a map-origin change.
Pose/velocity and supported uncertainty	Use finite, meaningful fields; covariance support is UNKNOWN.

WBR message, rate, transport and age threshold remain unselected.

WBR health supervision: semantics before firmware

KNOWN method / WBR qualification OPEN

- ABSENT and INITIALIZING supply no usable estimate; FRESH requires every eligibility check.
- Old age becomes STALE; bad fields, frame or reset become INVALID and withdraw optional input.
- A new session or map requires explicit re-establishment, not silent acceptance of its first packet.



Figure 26: Planned WBR semantics; OS INIT/FRESH/HOLD/FAILSAFE is a separate implementation.

Why WBR cares: a heartbeat proves liveness, not freshness or physical validity.

Health envelope: check yourself

SOURCE-SUPPORTED planned semantics

- Worked case: sequence advances and heartbeat is current, but pose epoch stays unchanged. Is the pose FRESH?

Work out your answer before turning the page.

Health envelope: answer

SOURCE-SUPPORTED planned semantics

- Not from that evidence. The receiver must bound source age; repeated transmission does not rejuvenate an old estimate.
- Test stale epochs, NaN, wrong frames, duplicates, clock wrap and resets in N6 fixtures before N7.

Explain the evidence that would distinguish the two cases.

N7: keep the control path independent

KNOWN method / WBR qualification OPEN

- The receiver may admit reviewed optional navigation data when its envelope is eligible.
- STM32 body IMU, local estimation, watchdog and control limits keep authority when the Pi disappears.
- First test a receiver bench without actuation, then separately approved controlled integration.



Figure 27: Independent local body sensing → balance → motor limits remains controller-owned.

Bench packets cannot establish physical balancing or a safe fallback time.

UART: worked serialization floors

CALCULATED from source framing

- WBR bench request/reply wire lengths imply ≈3.65 ms at 115200 baud.
- CN E7 sends chunked images; 30 kB plus framing is 30,510 UART bytes.
- At 115200 8N1, the E7 ingress line-time floor is ≈2.648 s.

$$T_{wire} = \frac{10\, N_{bytes}}{\text{baud}} \quad \text{(8N1)}$$

UART image ingress and small supervisory messages have different budgets.

CN/OS supervisor states and recovery

SOURCE-SUPPORTED OS integration / hardware UNTESTED

State	Source-defined behavior
INIT = 0	Readiness incomplete; fresh local and network evidence required.
FRESH = 1	All required liveness, age and validity checks healthy.
HOLD = 2	Results stale; can recover when all checks become healthy.
FAILSAFE = 3	Latched; board port requires reset, then new readiness.

Stop results with heartbeat alive to show HOLD before FAILSAFE.

Link and supervision: common mistake and check yourself

KNOWN method

- Mistake: stop the whole Pi bridge and expect HOLD→FAILSAFE.
- Local heartbeat loss triggers FAILSAFE at 100 ms; network stale-data path is separate.
- Question: do new packets automatically recover latched FAILSAFE?

Two fault injections test two mechanisms; record state and reason.

Link and supervision: answer

KNOWN method / WBR qualification OPEN

- No. New packets do not automatically clear latched FAILSAFE.
- HOLD can recover when checks become healthy; this OS board port requires reset for FAILSAFE, then new readiness.

Use this answer to check the reasoning on the preceding page.

Bottom-up diagnosis: find the earliest failed layer

KNOWN method / WBR qualification OPEN

- Start with power/cooling, own USB branch, access and raw stream identity.
- Then inspect clocks, wrapper graph/QoS, recording/replay and conversion fidelity.
- Only after calibration and input coverage pass should you inspect ORB state and trajectory metrics.

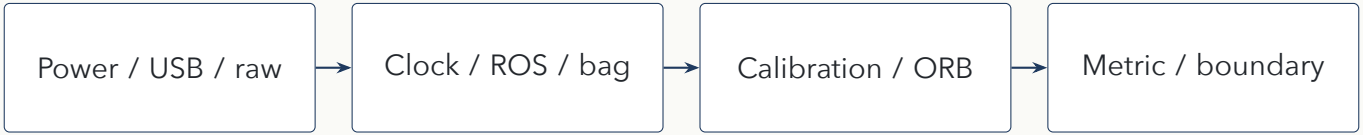


Figure 28: A failed earlier layer stops downstream acceptance; independent work can continue.

Why WBR cares: changing a late-stage parameter can conceal an upstream acquisition defect.

Diagnosis: symptom, evidence, next action

SOURCE-SUPPORTED contract / gate OPEN

Symptom	Inspect before changing anything
Images work, motion is silent	HID/IIO access, raw stream selection and exact topic graph.
Bag gaps appear only under recording load	Durable-write batches, mount identity and queue events.
Depth scale or gravity looks wrong	Frame direction, units, rectification and timestamps.
Offline works, live output ages	Queue wait, executor scheduling, CPU clock and loaded power.

Common mistake: reinstall or tune SLAM before preserving the discriminating evidence.

Bottom-up diagnosis: check yourself

KNOWN method / illustrative question

- A 15 s RAM take is clean but a card take develops 2 s gaps. Which next check separates hypotheses?

Work out your answer before turning the page.

Bottom-up diagnosis: answer

KNOWN method / illustrative question

- Inspect sustained durable-write batches and queue/backpressure logs under the same source load.
- The observation supports a storage-path hypothesis; it does not prove the sensor is faulty or the card is qualified.

Explain the evidence that would distinguish the two cases.

Engineering choices: what evidence would change them?

SOURCE-SUPPORTED contract / gate OPEN

Current choice	Evidence needed to reconsider
Native Pi runtime; direct USB3 cable	Equivalent physical gates plus a packaging/topology benefit.
Upstream EuRoC → offline camera → live	Matched-data equivalence before adding a wrapper or live path.
Measured calibration/noise before tuning	Import/model support and held-out residual improvement.
Defer wheel-odometry fusion	Synchronized geometry, covariance meaning and slip evidence.

Add complexity to address a measured failure; keep the comparison controlled.

Reproduction: the evidence packet is the output

KNOWN method / WBR qualification OPEN

- Keep raw data, device/boot identity, exact argv, settings and binary/library/source hashes.
- Link every derivative to its immutable inputs; preserve failures and actual review receipts.
- Separate an automatic capture result, a human review and a sealed acceptance decision.

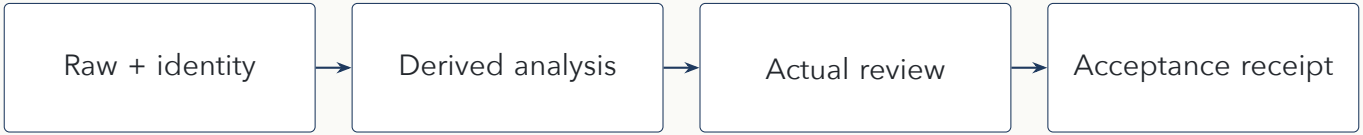


Figure 29: No prefilled approval and no editing deployed tools in place.

A complete packet lets another person recompute the claim without repeating the lab.

Bounded analysis tickets: one uncertainty at a time

KNOWN workflow / future execution OPEN

- State one question, allowed files/hosts/actions and a fixed evidence input.
- Require a discriminating check, retained outputs and explicit STOP conditions.
- The owner accepts gate and safety decisions; an analysis tool cannot manufacture missing hardware evidence.

> Example ticket: audit one take's timestamp order and explain every rejection.

Historical evidence: keep unit and date attached

MEASURED archived / all gates OPEN

- September x86/VM USB2 and later Pi 5000M→480M behavior explain transport risks.
- A historical 105 s capture did not complete N2; old-camera evidence does not transfer to this unit.
- Phase 8 adds the 8 GB boot, card gaps, clean short RAM take and interrupted crash-repair rebuild.

Historical progress explains the next test; it never silently becomes current acceptance.

Current gaps: what would close each one?

OPEN / dated source synthesis

Gap	Next evidence
N0–N3 platform/new-unit input	Loaded healthy power, qualified storage, fresh strict receipts.
N4A known dataset	Stable-power repeats and reviewed accuracy/continuity budgets.
N4B/C camera estimation	Stereo/time/calibration audit, motion replay, then live counters.
N5–N7 measurement / boundary	Raw event series, semantic fixtures, approved controller traces.

> The active project gate remains N3; diagnostic work does not replace it.

Conditional roadmap: proceed on evidence

SOURCE-SUPPORTED conditional plan / no gate PASS

- Refresh stable N0, renew N1 and finish exact N2/N3 before qualified camera claims.
- Phase-8 diagnostic roadmap: complete the repair build, record a 100 s RAM motion take, replay/plot, then live diagnostic.
- The later source reports only a clean still replay without a map; motion and live remain OPEN.

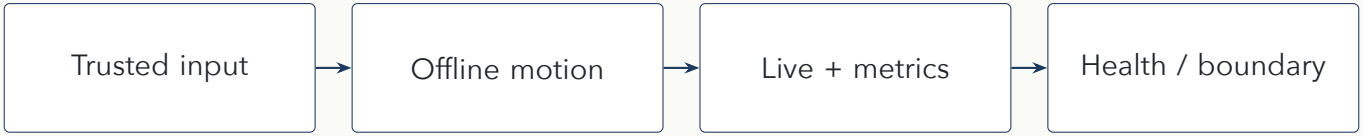


Figure 30: Dates and a successful launch do not advance a gate.

N4A can proceed independently; N6 review and N7 authorization remain separate.

Conclusion: explain the system aloud

KNOWN method / WBR qualification OPEN

- Describe how paired pixels and covered raw IMU become a pose in a named frame and epoch.
- Explain why geometry, noise, clocks, queues and storage each need separate evidence.
- Describe how stale, invalid or reset perception is withdrawn while STM32 retains local control.

> Understanding means predicting both the useful output and the failure path.

Lecture recap: check yourself

KNOWN method / illustrative question

- You have a plausible trajectory, clean exit and a live heartbeat. Which three checks still guard a receiver?

Work out your answer before turning the page.

Lecture recap: answer

KNOWN method / illustrative question

- Check frame/session/reset validity, finite supported state and a conservative source-age bound.
- Those checks complement acquisition and calibration evidence; a plausible path is only one part of the chain.

> Explain the evidence that would distinguish the two cases.

Lab operating contract

ASSUMED time budget / SOURCE-SUPPORTED stop rules

- Each supervised session is capped at 150 min; first failure consumes the retry buffer.
- No build, install, card clone or persistent network change belongs in these sessions.
- The 30 min warm-up and Allan take ($\geq$4 h provisional; 15–24 h preferred) happen at home.

> Stop at the time cap; leave skipped work OPEN.

Packing checklist

SOURCE-SUPPORTED packing / availability to verify

- 8 GB Pi + qualified 128 GB card + 27 W supply; 4 GB Pi + own supply for CN.
- D435i/cable, Ethernet, Mac/power, rigid measured AprilGrid, ruler/calipers, tape.
- F446RE, Mac USB cable, 3 jumpers; optional analyzer; prepared firmware and tools.

Pack distinct link-bench and OS-supervisor firmware with hashes.

Stop rules: the earliest failed layer wins

SOURCE-SUPPORTED prerequisites / ASSUMED one-retry policy

- New live power fault, reboot, USB <5000M or data reversal: stop acquisition.
- At a missing tool, unknown device identity or unverified command: stop that branch.
- One corrected retry with a new ID; then defer to Session 2 or home preparation.

> Do not tune SLAM, erase failed takes or relax thresholds to hide sensor faults.

Shell variables used in the lab

SOURCE-SUPPORTED shell bindings / chosen deployment paths

Variable / binding	Plain meaning
$HOME	The shell's existing home folder; never change it.
$GATE	Prepared complete gate bundle: ~/wbr-runtime/study-gates.
$RUN	This session's small logs/receipts: ~/wbr-evidence/study-S1-20261018.
$DATA	This session's take root: ~/wbr-takes/study-S1-20261018.
$STUDY	Prepared copy of this package: ~/wbr-runtime/study-docs.

Repeat bindings in each named terminal; use fresh actual session and take IDs.

Shell variables: take and terminal identity

SOURCE-SUPPORTED shell bindings / chosen deployment paths

Variable / binding	Plain meaning
$ID	Fresh take name, such as S-ROS-01 or C1-ROS-01; repeat in every terminal.
$TOPICS	Bash array of explicit topics; define in the recorder shell before use.
$TAKE	Completed native take name used by the handoff copy loop.
$PORT	Actual checked UART device, such as /dev/ttyAMA0; not a guessed port.
$WBR_HOST	Prepared demo SSH target richard@192.168.50.2; verify against inventory.

Repeat bindings in each named terminal; use fresh actual session and take IDs.

Shell variables: home analysis paths

SOURCE-SUPPORTED shell bindings / chosen deployment paths

Variable / binding	Plain meaning
$ARCH	Copied Mac session root under ~/Desktop/Research/wbr-data/.
$BAG	An actual finalized ROS bag directory, not a native SDK file.
$NATIVE	An actual finalized native take/ folder containing stereo_imu.db3.
$DERIVED	Fresh output parent for analysis; keep it separate from raw takes.
$TARGET	Measured target YAML at the explicit H4 path; nominal dimensions are insufficient.

Repeat bindings in each named terminal; use fresh actual session and take IDs.

Shell variables: ORB and CN bindings

SOURCE-SUPPORTED shell bindings / chosen deployment paths

Variable / binding	Plain meaning
$ORB_DIR / $TAKE_DIR	Prepared ORB root / audited exported sequence used in H6.
$SETTINGS_FILE / $LD_LIBRARY_PATH	Reviewed ORB YAML / runtime library search list from H6.
$WIFI_B	Pi B's measured Wi-Fi IPv4 address; Pi A must route to it over wlan0.
$CN_CLIENT_SSH / $CN_EDGE_SSH	Actual client/edge user@host bindings entered during S2 archive.

Repeat bindings in each named terminal; use fresh actual session and take IDs.

Shell variables: repeat bindings in each terminal

SOURCE-SUPPORTED shell bindings / chosen deployment paths

Variable / binding	Plain meaning
Session 2 RUN / DATA	Use study-S2-20261019 as shown in S2.0; dates are examples, not observations.
ROS_DOMAIN_ID=42 / 43	Environment: localhost live acquisition / isolated home replay.
ROS_LOCALHOST_ONLY=1	Environment: keep the ROS graph on the current machine.
STUDY versus ~/CN	STUDY holds study configs; ~/CN is the separately prepared CN checkout.

Repeat bindings in each named terminal; use fresh actual session and take IDs.

Session 1 map: stop at the earliest failed prerequisite

ASSUMED session decision map

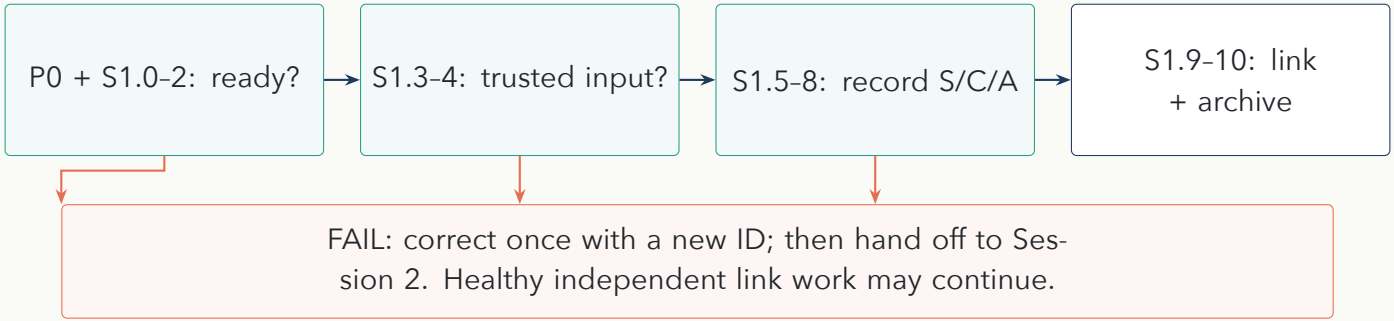


Figure 31: YES follows the top arrows; failed readiness follows the lower branch. Stop at 150 minutes.

Read the step card before its commands. Preserve failed and skipped work as evidence.

Session 1 timeline: 150 min maximum

ASSUMED wall-time budget

Elapsed	Task / exit artifact
0–15 / 15–30 min	S1.0–2: setup, power, storage/USB identity.
30–55 / 55–75 min	S1.3–4: N2 stages/review; N3 audit/record-replay.
75–110 / 110–125 min	S1.5–7: ROS S/C takes; native replicas only if ready.
125–140 / 140–150 min	S1.8–10: A pilot plan, link bench, verified handoff.

N3 unresolved at minute 75 means calibration qualification is deferred.

Before lab: prepare, do not improvise

SOURCE-SUPPORTED runbook / readiness OPEN

- Owner-authorized transfer/deployment, card setup and any builds happen beforehand.
- Gate bundle, wrapper, ROS CLI, native recorder, Kalibr image and CN files must exist.
- The strict card exception needs a reviewed sustained-write receipt; capacity is insufficient.

> An unready prerequisite becomes a documented skipped step.

P0 | Bind prepared paths and verify tools

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Check the prepared tools and name the output folders before connecting hardware, so every later command has a known location.

Physical setup: On the Mac, open the canonical WBR checkout. On each Pi terminal, source Jazzy and repeat the path bindings. Keep all camera programs stopped.

What you see: The help command lists the demo interface. Each file test returns without an error; the new RUN and DATA folders exist.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Continue only if every required file is present and the chosen output folders are fresh. A missing file leaves this branch OPEN.

If FAIL: Stop this branch and prepare the missing dependency at home. Do not install tools or guess another script during the session.

Take these files home: Keep Pi logs in RUN and takes in DATA. Keep the source revision and dirty-file record with the Mac session archive.

P0 | Mac / before lab (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/Desktop/Research/wbr-perception-phase1-runtime
export WBR_HOST=richard@192.168.50.2
bash experiments/2026-10-15-demo-day/demo.sh --help
git rev-parse HEAD
git status --short
```

> Use the checks and failure branch on the preceding P0 step card.

P0 | Pi / each shell (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
source /opt/ros/jazzy/setup.bash
export GATE="$HOME/wbr-runtime/study-gates"
export RUN="$HOME/wbr-evidence/study-S1-20261018"
export DATA="$HOME/wbr-takes/study-S1-20261018"
# Prepared, reviewed 128 GB data-directory exception.
mkdir -p "$RUN" "$DATA"
test -f "$GATE/phase_gate.py"
test -f "$GATE/n3_record.py"
test -f "$GATE/n3_bag.py"
export ROS_DOMAIN_ID=42 ROS_LOCALHOST_ONLY=1
```

> Use the checks and failure branch on the preceding P0 step card.

S1.0 | Set up and make the evidence manifest

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Document the physical assembly before testing it, so a result can be traced to the board, supply, cable and storage actually used.

Physical setup: Secure the Pi and camera cable against pulling. Use the direct USB3 cable and connect power last. Camera access requires the owner's CAMERA AVAILABLE instruction.

What you see: The identity file shows the running kernel and resolved data mount. Your notebook identifies the supply, cooler, board, card, target and room.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Continue when every identity field is recorded or explicitly marked UNKNOWN, and the process list shows no competing camera owner.

If FAIL: Correct cabling or process ownership first. Record any missing identity; use a fresh run folder rather than overwriting earlier evidence.

Take these files home: Save RUN/identity.txt and processes-before.txt. Retain the Mac source/config hash files and measured-target notes in the same session archive.

SOURCE-SUPPORTED procedure / execution OPEN

```
uptime > "$RUN/identity.txt"
uname -a >> "$RUN/identity.txt"
findmnt -T "$DATA" >> "$RUN/identity.txt"
lsblk -o NAME,SIZE,MODEL,FSTYPE,UUID,MOUNTPOINTS \
  >> "$RUN/identity.txt"
ps -eo pid,args > "$RUN/processes-before.txt"
```

Use the checks and failure branch on the preceding S1.0 step card.

S1.0 | Mac / canonical checkout (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
git rev-parse HEAD > /tmp/study-source-git.txt
git status --short > /tmp/study-source-dirty.txt
shasum -a 256 calibration/lab/config/*.yaml \
  calibration/aprilgrid/*.yaml > /tmp/study-config.sha256
```

> Use the checks and failure branch on the preceding S1.0 step card.

S1.1 | Power and storage preflight

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Rule out the two observed bottlenecks, weak power and slow storage, before collecting new sensor data.

Physical setup: Use the 8 GB Pi with its 27 W supply and cooling. Point DATA at the prepared, qualified 128 GB data directory and have its write receipt available.

What you see: The commands print raw power flags, rail voltage, temperature, available memory and disk space. A healthy power example is throttled=0x0.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: N0 requires flags 0x0, temperature below 75°C and available memory ≥512 MiB. The card exception proposes sustained writes ≥27.6 MB/s; this margin is ASSUMED.

If FAIL: Stop acquisition after a power fault or reboot. Check supply, cable and cooling, then storage identity and write evidence. An unqualified volume cannot receive a qualified take.

Take these files home: Keep RUN/power-before.txt and capacity.txt, the original write receipt and its hash. Record the actual free-space reserve before each take.

S1.1 | Pi / control (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
vcgencmd get_throttled | tee "$RUN/power-before.txt"
vcgencmd pmic_read_adc EXT5V_V \
  | tee -a "$RUN/power-before.txt"
vcgencmd measure_temp | tee -a "$RUN/power-before.txt"
free -h | tee "$RUN/capacity.txt"
df -B1 "$DATA" /dev/shm | tee -a "$RUN/capacity.txt"
timedatectl status > "$RUN/time-before.txt"
```

> Use the checks and failure branch on the preceding S1.1 step card.

S1.2 | USB, serial and fresh N1

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Prove access to the current camera on its own USB branch before asking it to stream.

Physical setup: Connect the D435i directly to the Pi's blue USB3 port. Use stable N0 power, the complete prepared GATE bundle and the owner's camera authorization.

What you see: The camera's own lsusb branch shows at least 5000M. SDK enumeration identifies serial 317422076650 and reports firmware and the other identity fields separately.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Continue only if the camera branch is ≥5000M and n1_precheck exits 0 with an eligible sealed receipt. Access or kernel errors fail this check.

If FAIL: At 480M, stop before SDK enumeration. Inspect cable, port and intervening topology in that order; retry once into a new output folder.

Take these files home: Keep RUN/usb.txt, sdk-enumeration.txt and the entire RUN/n1 directory, including result.json and its sealed raw logs.

S1.2 | Pi / control (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
lsusb -t | tee "$RUN/usb.txt"
# Only after camera branch is confirmed >=5000M:
rs-enumerate-devices > "$RUN/sdk-enumeration.txt" 2>&1
cd "$GATE"
python3 n1_precheck.py --camera-available \
  --direct-cable-confirmed --output "$RUN/n1"
```

Use the checks and failure branch on the preceding S1.2 step card.

S1.3 | Exact N2 acquisition: inspect each stage

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Prove sustained raw delivery before adding ROS or SLAM; this isolates the sensor path from later software layers.

Physical setup: Keep only the native gate recorder running. Use a fresh eligible N1 receipt and the exact IR1 640×480 Y8/15, gyro/200 and accel/100 profile.

What you see: Run 20 s, inspect the result, then 105 s, inspect again, then the mandatory 300 s soak. Each stage writes raw CSV, image samples and one-Hz health.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require IR 14.25–15.75, gyro 190–210 and accel 95–105 Hz. Image gaps must be $\leq$0.20 s and motion gaps $\leq$0.05 s, with positive steps and no missing IR frames.

If FAIL: Stop at the first failure. Inspect power, USB, permissions, selected profile and timestamps in that order. Allow one corrected new-ID retry, then defer to Session 2.

Take these files home: Preserve RUN/smoke, reproduction and soak in full, including capture-result.json, CSV files, SDK/kernel logs and sampled PGMs.

S1.3 | Pi / gate bundle, stage 1 (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd "$GATE"
python3 phase_gate.py n2 --camera-available --seconds 20 \
  --n1-result "$RUN/n1/result.json" \
  --output "$RUN/smoke"
```

> Use the checks and failure branch on the preceding S1.3 step card.

S1.3 | Pi / only after prior stage review (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 phase_gate.py n2 --camera-available --seconds 105 \
  --n1-result "$RUN/n1/result.json" \
  --previous "$RUN/smoke/capture-result.json" \
  --output "$RUN/reproduction"
# Review that result before issuing the next command.
python3 phase_gate.py n2 --camera-available --seconds 300 \
  --n1-result "$RUN/n1/result.json" \
  --previous "$RUN/reproduction/capture-result.json" \
  --output "$RUN/soak"
```

> Use the checks and failure branch on the preceding S1.3 step card.

S1.3R | Human N2 review and acceptance

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Turn completed captures into an actual reviewed decision; an automatic zero exit is not human acceptance.

Physical setup: Open the sampled images and unfiltered logs from all three strict N2 stages. Have the real reviewer identity, UTC time and observations ready.

What you see: The review file records your actual observations and retained warnings. The acceptance command checks the sealed evidence instead of trusting a claim of success.

S1.3R | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Accept only coherent images, finite plausible motion, explained startup offsets and preserved warnings, followed by eligible acceptance output. Incomplete review leaves N2 OPEN.

If FAIL: Keep the gate OPEN when any required review is missing. Do not prefill true booleans or replace a strict receipt with a diagnostic capture.

Take these files home: Save RUN/review.json and RUN/n2-accepted/result.json beside all original stage directories.

S1.3R | Pi / after actual review (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 phase_gate.py accept-n2 --camera-available \
  --n1-result "$RUN/n1/result.json" \
  --smoke "$RUN/smoke/capture-result.json" \
  --reproduction "$RUN/reproduction/capture-result.json" \
  --soak "$RUN/soak/capture-result.json" \
  --review "$RUN/review.json" --output "$RUN/n2-accepted"
```

> Use the checks and failure branch on the preceding S1.3R step card.

S1.4 | Active N3: baseline record and replay

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Check that ROS publishes the expected data and preserves it through recording and replay before making ROS-based calibration claims.

Physical setup: Use reviewed N2, the installed reviewed wrapper and a storage receipt the strict runner accepts. Stop the live camera before the replay phase.

What you see: The runner observes 60 s live, then records a 20 s sqlite3 bag. Replay uses localhost domain 43 while the live domain remains isolated.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require the exact baseline topics/types, N2 rate bands and complete replay agreement in counts, headers and payloads. Live contamination fails the check.

If FAIL: If the runner cannot accept the approved card policy or receipt, stop and leave N3 OPEN. Otherwise inspect raw delivery, graph/QoS and storage before retrying once.

Take these files home: Retain DATA/N3 plus RUN/n3-record and n3-replay, including parameter readback, observer/player logs, metadata and result files.

S1.4 | Pi / prepared bundle (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 wrapper_inspect.py --output "$RUN/wrapper.json"
# UNVERIFIED: card-exception receipt/runner compatibility.
# Run only with a reviewed eligible storage receipt.
python3 n3_record.py --camera-available \
  --n1-result "$RUN/n1/result.json" \
  --n2-result "$RUN/n2-accepted/result.json" \
  --storage-result "$RUN/storage/result.json" \
  --bag "$DATA/N3" --seconds 20 --output "$RUN/n3-record"
```

> Use the checks and failure branch on the preceding S1.4 step card.

S1.4 | Pi / after live camera has stopped (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 n3_bag.py replay "$DATA/N3" \
  --record-result "$RUN/n3-record/result.json" \
  --live-domain 42 --replay-domain 43 \
  --output-dir "$RUN/n3-replay"
```

> Use the checks and failure branch on the preceding S1.4 step card.

Record contract: one device owner, explicit representations

SOURCE-SUPPORTED ownership / conversion gaps OPEN

- Main S/C/A takes use one ROS wrapper plus ros2 bag record, explicit topics and sqlite3.
- Native S/C replicas use the checked recorder after ROS stops; their own IDs stay distinct.
- Native playback→ROS recording is source-supported but target compatibility is UNVERIFIED.

> Native Allan CSV has no supplied ROS companion adapter; that branch is blocked.

S1.5 | ROS recipe shared by S, C1–C3 and A

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Use one explicit ROS recording recipe for S and C takes, so the sensor data and the context needed at home are preserved together.

Physical setup: Copy this study package to STUDY before lab. Use one wrapper, three named terminals and the same RUN, DATA and ID in each. Review actual graph and option readback first.

What you see: The graph contains /camera/camera streams and metadata. The recorder creates DATA/ID/ros while the other terminals save parameters, wrapper output and periodic power samples.

S1.5 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: For the ASSUMED 30 Hz review, images must be 28.5–31.5 Hz; gyro 190–210 and accel 95–105 Hz. Reject reversals, skipped samples or gaps >1.5 nominal periods.

If FAIL: Stop if metadata or clock domains are missing. Inspect power, USB, profile, QoS and storage in order; make at most one corrected attempt with a new ID.

Take these files home: Preserve DATA/ID/ros and RUN/ID together: parameters, topics, QoS, wrapper.log, bag-info and all power samples. A takes use the separate S1.8R topic recipe.

S1.5 | Pi / camera terminal A (1/5)

SOURCE-SUPPORTED procedure / execution OPEN

```
export STUDY="$HOME/wbr-runtime/study-docs"
: "${ID:?Bind this step ID in each Pi terminal first}"
mkdir -p "$RUN/$ID"
set -o pipefail
# Source-checked recipe; installed wrapper remains UNVERIFIED.
ros2 launch realsense2_camera rs_launch.py \
  config_file:="$STUDY/lab-config/ros-stereo.yaml" \
  2>&1 | tee "$RUN/$ID/wrapper.log"
# Keep this terminal open. Ctrl-C after the bag finalizes.
```

Use the checks and failure branch on the preceding S1.5 step card.

SOURCE-SUPPORTED procedure / execution OPEN

```
: "${ID:?Set current step ID, for example S-ROS-01}"
mkdir -p "$RUN/$ID" "$DATA/$ID"
ros2 param dump /camera/camera > "$RUN/$ID/params.yaml"
ros2 topic list -t > "$RUN/$ID/topics.txt"
ros2 topic info --verbose /camera/camera/gyro/sample \
  > "$RUN/$ID/gyro-qos.txt"
vcgencmd get_throttled > "$RUN/$ID/power-before.txt"
```

Use the checks and failure branch on the preceding S1.5 step card.

S1.5 | Pi / topic selection terminal B (3/5)

SOURCE-SUPPORTED procedure / execution OPEN

```
TOPICS=(
/camera/camera/infra1/image_rect_raw
/camera/camera/infra2/image_rect_raw
/camera/camera/infra1/camera_info /camera/camera/infra2/camera_info
/camera/camera/gyro/sample /camera/camera/accel/sample
/camera/camera/gyro/imu_info /camera/camera/accel/imu_info
/camera/camera/extrinsics/depth_to_infra1
/camera/camera/extrinsics/depth_to_infra2
/camera/camera/extrinsics/depth_to_gyro
/camera/camera/extrinsics/depth_to_accel /tf_static
/camera/camera/infra1/metadata /camera/camera/infra2/metadata
/camera/camera/gyro/metadata /camera/camera/accel/metadata
)
```

> Use the checks and failure branch on the preceding S1.5 step card.

S1.5 | Pi / recorder terminal B, same shell (4/5)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED: installed metadata presence/field semantics.
ros2 bag record --storage sqlite3 --output "$DATA/$ID/ros" \
  --qos-profile-overrides-path "$STUDY/lab-config/qos.yaml" \
  --topics "${TOPICS[@]}"
# Ctrl-C after this take's duration; wait for finalization.
ros2 bag info "$DATA/$ID/ros" > "$RUN/$ID/bag-info.txt"
vcgencmd get_throttled > "$RUN/$ID/power-after.txt"
```

Use the checks and failure branch on the preceding S1.5 step card.

S1.5 | Pi / telemetry terminal C, during each take (5/5)

SOURCE-SUPPORTED procedure / execution OPEN

```
# Bind RUN and ID as in terminal B; Ctrl-C after capture stops.
while true; do
  date -u +%FT%TZ
  vcgencmd get_throttled
  vcgencmd pmic_read_adc EXT5V_V
  vcgencmd measure_temp
  sleep 1
done > "$RUN/$ID/power-periodic.txt" 2>&1
```

Use the checks and failure branch on the preceding S1.5 step card.

S1.5F | Finalize, hash and preserve each ROS take

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Finalize and verify each take before copying it, so home analysis reads a complete bag rather than files still being written.

Physical setup: Press Ctrl-C in the recorder and wait for metadata finalization. Then stop the wrapper and telemetry. No process may still write either take tree.

What you see: ros2 bag info opens the bag and lists the selected topics. The local and copied SHA-256 checks report that the finalized files match.

S1.5F | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require every selected topic with nonzero expected counts and matching Mac hashes. This proves preservation, not the sensor rates or calibration quality.

If FAIL: Keep the Pi originals. Retry a failed copy once; a mismatch or missing context leaves the take unusable for qualification until resolved.

Take these files home: Copy DATA/ID and RUN/ID to the Mac session archive. Keep metadata.yaml, every db3 file, SHA256SUMS, configs and the failed attempts.

S1.5F | Pi / after finalization (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
cp "$STUDY/lab-config/ros-stereo.yaml" "$RUN/$ID/"
cp "$STUDY/lab-config/qos.yaml" "$RUN/$ID/"
cd "$DATA/$ID"
find ros -type f -print0 | sort -z | xargs -0 sha256sum \
  > SHA256SUMS
sha256sum -c SHA256SUMS
# Save periodic telemetry from a separate terminal during takes.
# For all S/C/A takes: before, during, after; errors stay visible.
```

Use the checks and failure branch on the preceding S1.5F step card.

S1.5F | Mac / after all writers stop (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
mkdir -p ~/Desktop/Research/wbr-data/study-S1-20261018
rsync -a richard@192.168.50.2:wbr-takes/study-S1-20261018/ \
  ~/Desktop/Research/wbr-data/study-S1-20261018/
cd ~/Desktop/Research/wbr-data/study-S1-20261018/S-ROS-01
shasum -a 256 -c SHA256SUMS
```

Use the checks and failure branch on the preceding S1.5F step card.

S1.6 | S stereo take: static intervals then coverage

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Record S to validate factory stereo geometry before interpreting disparity as metric depth.

Physical setup: Fix the measured 6×4 grid to a flat board. Turn the emitter off, use suitable near-infrared light, and move the camera smoothly without cable pull.

What you see: Record 120 s: hold still at 0.5, 1 and 1.5 m for 10 s each, then sweep and tilt slowly. If far tags do not decode in both views, use the stated visibility fallback.

S1.6 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Home review proposes ≥100 paired detections in four quadrants, RMS ≤0.5 px, row median ≤0.3 px/P95 ≤1 px and depth error ≤2%. These are ASSUMED thresholds.

If FAIL: Inspect print size/flatness, light/exposure, pairing and timestamps before changing calibration. Mark invisible far ranges OPEN and retry a failed take in Session 2.

Take these files home: Keep DATA/S-ROS-01/ros, RUN/S-ROS-01 and target/range measurements. An optional native replica has a different ID and remains blocked until its companion route is validated.

S1.6 | Pi / ROS control before S1.5 (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ID=S-ROS-01
# Run S1.5 with this ID; record 120 seconds.
# 0-10s: still at 0.5m; 10-20s: 1m; 20-30s: 1.5m.
# Use those ranges only if the small grid decodes in both IRs.
# If not: 0.4 / 0.6 / 0.8m visibility run, mark far-depth OPEN.
# 30-120s: slow sweeps / tilts; four-quadrant coverage.
# Finalize and hash with S1.5F before continuing.
```

Use the checks and failure branch on the preceding S1.6 step card.

S1.6 | Pi / OPTIONAL native replica, ROS stopped (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export WBR_CAMERA_AVAILABLE='CAMERA AVAILABLE'
# Native replica is a new take; repeat the S motion recipe.
bash ~/wbr-runtime/calib-lab-20261015/pi/record_calib_take.sh \
  S-NATIVE-01 120
# Requires 3,974,400,000 free tmpfs bytes plus disk reserves.
# UNVERIFIED: native playback -> derived ROS companion (H2).
# Without a validated companion route, defer native replica.
```

> Use the checks and failure branch on the preceding S1.6 step card.

S1.7 | C1–C3 camera–IMU takes

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Record three independent motions so home fitting can test whether the camera–IMU transform and time shift repeat.

Physical setup: Keep the measured target fixed and visible in both IR views. Preserve settings and domains. Move the camera through roll, pitch, yaw and XYZ translation without impacts.

What you see: Each 60 s C take starts and ends still. Between these intervals, use smooth rotations and ±15–20 cm translations; retain the whole target and avoid blur.

S1.7 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Capture must have allowed known domains and no gaps, skips, blur or saturation. Home review proposes $\leq$3 mm/1° extrinsic spread and $\leq$1 ms shift spread across three takes.

If FAIL: Correct one visibility or motion problem, then retry once with a new ID. Defer missing takes to Session 2; never reuse a failed take's folder.

Take these files home: Keep DATA/C1-ROS-01/ros through C3 and their RUN logs. Optional C-native replicas are separately named physical takes, not equivalent copies of the ROS takes.

S1.7 | Pi / ROS control, each take separately (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ID=C1-ROS-01
# Run S1.5 for 60 seconds, then S1.5F.
# Repeat with ID=C2-ROS-01, then ID=C3-ROS-01.
# 0-10s still; 10-25s rotations about each axis.
# 25-45s XYZ translations, +/-15-20cm, mixed motion.
# 45-50s varied target position/range; 50-60s still.
# A3: 0.4-0.8m; A4: 0.3-0.5m. No taps/cable pull.
```

> Use the checks and failure branch on the preceding S1.7 step card.

S1.7 | Pi / OPTIONAL native replicas, ROS stopped (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export WBR_CAMERA_AVAILABLE='CAMERA AVAILABLE'
bash ~/wbr-runtime/calib-lab-20261015/pi/record_calib_take.sh \
  C1-NATIVE-01 60
# Repeat recipe and command as C2-NATIVE-01 / C3-NATIVE-01.
# Each 60s native take needs >=1,987,200,000 tmpfs bytes.
# UNVERIFIED: H2 native-to-ROS companion round trip.
# Defer replicas when that route is not validated.
```

> Use the checks and failure branch on the preceding S1.7 step card.

S1.8 | Allan pilot and between-session long recording

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Use a short pilot to inspect raw timing, then reserve hours at home for noise processes that cannot be identified in a short lab visit.

Physical setup: Rest the camera on a rigid vibration-free surface with cable strain relief. Use stable power/storage and IMU-only recording; warm up 30 min before the home long run.

What you see: The lab pilot lasts 36 s. After review, the home plan records 15 h; 4 h is provisional and 15–24 h is preferred. No image traffic is requested.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: The ASSUMED capture review requires raw rates within ±5%, monotonic allowed domains and no missing samples. Document stationarity and temperature/power evidence.

If FAIL: Stop on power faults. Never run ROS and native owners together. The optional native CSV branch stays BLOCKED while the required CSV-to-ROS companion is absent.

Take these files home: Keep A-ROS-pilot-01/ros and the separate home A-ROS-15h-01/ros with all logs. Preserve native allan-* directories only if that blocked branch becomes ready.

S1.8 | Pi / ROS camera terminal, replace stereo launch (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ID=A-ROS-pilot-01
mkdir -p "$RUN/$ID"
set -o pipefail
ros2 launch realsense2_camera rs_launch.py \
  config_file:="$STUDY/lab-config/ros-imu.yaml" \
  2>&1 | tee "$RUN/$ID/wrapper.log"
# A separate recorder terminal runs S1.8R, not S1.5 stereo.
# 36s pilot; clean stop, review, warm 30min before long take.
# Plan 15h at home (4h provisional; target 15-24h).
# Run at home only with maintained safe power/storage setup.
```

Use the checks and failure branch on the preceding S1.8 step card.

S1.8 | Pi / native alternative, BLOCKED pending companion (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED: no supplied raw-CSV -> ROS2 companion adapter.
# Do not execute this branch until that requirement is resolved.
cd ~/wbr-runtime/allan-20261014
bash ./record_allan.sh 0.01 A-native-pilot-01
# After reviewed pilot and 30min warm-up, at home only:
nohup bash ./record_allan.sh 15 A-native-15h-01 \
  > "$HOME/allan-A-native-15h-01.launch.log" 2>&1 < /dev/null &
```

Use the checks and failure branch on the preceding S1.8 step card.

S1.8R | Allan ROS recording: explicit IMU-only bag

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Preserve every primary A take as an explicit ROS 2 bag while recording only the raw IMU and required context.

Physical setup: Keep the IMU-only wrapper from S1.8 running. In the recorder terminal, repeat the same RUN, DATA, STUDY and ID; monitor power in another terminal.

What you see: The sqlite3 bag contains raw gyro and accel, their metadata and context, and /tf_static. No image topic appears in the selected recording list.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require the expected raw-rate counts, finite values, source-domain metadata and clean finalization. The 36 s pilot is not accepted as long-term noise evidence.

If FAIL: If metadata is missing or data are lost, retain the bag and reject qualification. Review the raw cause at home before starting a long replacement run.

Take these files home: Keep DATA/A-ROS-pilot-01/ros and RUN/A-ROS-pilot-01. Use the same structure for the separately named 15 h home take and copy the IMU-only config.

S1.8R | Pi / recorder terminal B (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ID=A-ROS-pilot-01
mkdir -p "$RUN/$ID" "$DATA/$ID"
ros2 param dump /camera/camera > "$RUN/$ID/params.yaml"
ros2 bag record --storage sqlite3 --output "$DATA/$ID/ros" \
  --qos-profile-overrides-path "$STUDY/lab-config/qos.yaml" \
  --topics /camera/camera/gyro/sample \
  /camera/camera/accel/sample /camera/camera/gyro/metadata \
  /camera/camera/accel/metadata \
  /camera/camera/gyro/imu_info /camera/camera/accel/imu_info \
  /camera/camera/extrinsics/depth_to_gyro \
  /camera/camera/extrinsics/depth_to_accel /tf_static
# Ctrl-C after 36s pilot; at home use ID=A-ROS-15h-01.
# Finalize and hash with S1.5F, copy ros-imu.yaml as config.
```

> Use the checks and failure branch on the preceding S1.8R step card.

S1.9 | Step 9b link bench: wire and check

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Test UART packet integrity with camera load removed before interpreting any higher-level supervisor behavior.

Physical setup: Unplug the camera. The Mac powers the F446RE with the prepared bench image. Cross Pi TX pin 8 to D2/PA10 and RX pin 10 to D8/PA9; join GND pin 6.

What you see: Preflight receives about 20 status records/s in NO_LINK. The LED follows packet traffic; this firmware contains no motor or balancing control.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Source expectations are 3000 sent/0 lost, CRC/duplicate/session rejection 10/10 each, and zero wire errors or MCU reboots. Keep the diagnostic label.

If FAIL: Check power, ground, crossed pins, port/console and firmware in order. If ttyAMA0 is absent, defer owner setup or use the documented USB fallback; never alter timeouts.

Take these files home: Keep the entire /tmp/wbr_link_bench/out-ID directory and firmware hash in the Mac session/link-bench archive before temporary Pi files disappear.

S1.9 | Mac / before wiring, prepared binary (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/Desktop/Research/wbr-perception-phase1-runtime
test -f tools/link_bench/firmware/prebuilt/link_bench_f446.bin
shasum -a 256 \
  tools/link_bench/firmware/prebuilt/link_bench_f446.bin
# Owner flashes this bench image from Mac before the session.
# Never flash the CN/OS image for this test or power MCU from Pi.
```

> Use the checks and failure branch on the preceding S1.9 step card.

S1.9 | Pi / control, camera unplugged (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
ls -l /dev/ttyAMA* /dev/ttyACM* /dev/serial*
groups
grep -n uart /boot/firmware/config.txt
cat /proc/cmdline
# Verify no console/getty owns the chosen device.
export PORT=/dev/ttyAMA0
python3 /tmp/wbr_link_bench/link_bench.py \
  --port "$PORT" --out /tmp/wbr_link_bench/out-S1-01
```

Use the checks and failure branch on the preceding S1.9 step card.

S1.9T | Step 9b timeouts and optional 9c

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Verify the bench silence rule, and optionally separate wire timing from software delay with a logic analyzer.

Physical setup: Finish 9b first and leave the camera unplugged. For optional 9c, use the documented 3.3 V probes, 12 MHz sampling and a 15 s capture.

What you see: The 100 ms silences should produce 0/5 NO_LINK events. The 300 ms silences should produce 5/5, around 150–151 ms MCU silence, then 5/5 recoveries.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require those source-defined observable counts. NO_LINK belongs to the bench protocol and does not establish OS HOLD/FAILSAFE or robot fallback.

If FAIL: Check baud, channels and ground before interpreting timing. If the Pi marker never toggles, report 9b-only timing; retry once, then defer.

Take these files home: Preserve the 9b output and, if captured, la-S1-01/summary.json, the original VCD/digital CSV and analyzed la.json.

S1.9T | Pi / only after analyzer capture starts (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 /tmp/wbr_link_bench/link_bench.py \
  --port "$PORT" --out /tmp/wbr_link_bench/la-S1-01 \
  --phases rtt --duration 10 --marker-gpio 17
```

Use the checks and failure branch on the preceding S1.9T step card.

S1.9T | Mac / after copying capture and summary (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/Desktop/Research/wbr-perception-phase1-runtime
python3 tools/link_bench/pi/la_analyze.py cap.vcd \
  --summary la-S1-01/summary.json --out la-S1-01/la.json
```

> Use the checks and failure branch on the preceding S1.9T step card.

S1.10 | Session handoff: stop at 150 minutes

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: End Session 1 within 150 min with preserved evidence and a short list of the earliest failed conditions to revisit.

Physical setup: Stop all writers normally, then copy and verify every completed directory on the Mac. Do not clone a card during this handoff.

What you see: handoff.md lists completed, failed, marginal and skipped steps, each with its evidence path, cause and required next action.

S1.10 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require matching hashes and a next action for every failure. A diagnostic result or unfinished review must not silently become a gate PASS.

If FAIL: Keep Pi originals when copying fails. Retry the transfer once; do not delete data or draw a supported-data conclusion from an incomplete archive.

Take these files home: Take home the full session archive, RUN logs, target notes and handoff.md, including Allan availability and the exact new IDs planned for Session 2.

S1.10 | Mac / copy small logs too (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
mkdir -p ~/Desktop/Research/wbr-data/study-S1-20261018/logs
rsync -a \
  richard@192.168.50.2:wbr-evidence/study-S1-20261018/ \
  ~/Desktop/Research/wbr-data/study-S1-20261018/logs/
# Review SHA256SUMS for every copied take as in S1.5F.
# Record A-ROS long-run availability and native/ROS gaps.
# Owner handles clean shutdown after evidence verification.
```

Use the checks and failure branch on the preceding S1.10 step card.

S1.10 | Mac / completed native or bench branches only (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ARCH="$HOME/Desktop/Research/wbr-data/study-S1-20261018"
export WBR_HOST=richard@192.168.50.2
# For EACH completed native ID only; example C1:
export TAKE=C1-NATIVE-01
mkdir -p "$ARCH/native/$TAKE"
rsync -a "$WBR_HOST:wbr-evidence/2026-10-15-calib-$TAKE/" \
  "$ARCH/native/$TAKE/"
(cd "$ARCH/native/$TAKE/take" && \
  shasum -a 256 -c ../ram-sha256.txt)
# Repeat TAKE=S-NATIVE-01 / C2-NATIVE-01 / C3-NATIVE-01 if done.
# If 9b/9c ran, preserve its entire output before /tmp is cleared:
rsync -a "$WBR_HOST:/tmp/wbr_link_bench/" "$ARCH/link-bench/"
```

Use the checks and failure branch on the preceding S1.10 step card.

SOURCE-SUPPORTED inventory / optional branches conditional

Bag or file family	Home consumer
N1/N2/N3 receipts, CSV/PGMs and logs	H0: input and replay context; retain all failures.
S and C1–C3 ROS bags + metadata/configs	H0 audit; H3 missing exporter; H4 fits after conversion.
A pilot / separate long A + power logs	H0 audit; H3 raw extraction; H5 supported Allan.
Optional native S/C + raw timestamps	H1 export, H2 companion, H3 conversion, H4 fits.

Each ROS bag means metadata.yaml plus every db3 file, finalized and hashed.

Session 1: what you take home — supporting files

SOURCE-SUPPORTED inventory / missing routes remain OPEN

Bag or file family	Home consumer
RUN/ID wrapper, params, topics, QoS, power	H0/H3 conversion checks; H4–H6 provenance.
Target YAML/dimensions, ranges and motion notes	H4 geometry; H7 proxy/reference interpretation.
Native A CSV/meta, only if unblocked	H3 preservation receipt, then H5 Allan fitting.
Link summary/raw/optional VCD; handoff; hashes	H0 archive check; retain link review separately.

H6 needs an actual motion export; H7 needs an independent reference for ATE/RPE.

Session 2 map: keep independent branches separate

ASSUMED session decision map

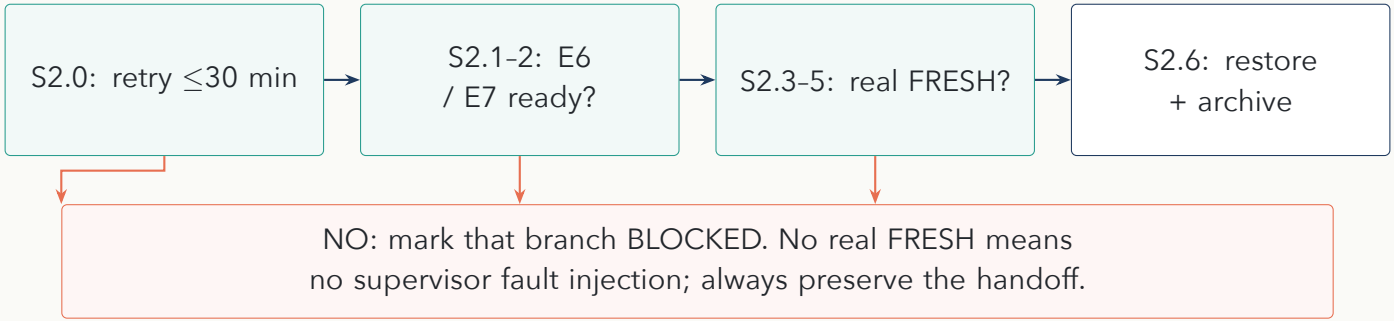


Figure 32: YES follows the top arrows; failed readiness follows the lower branch. Stop at 150 minutes.

Read the step card before its commands. Preserve failed and skipped work as evidence.

Session 2 timeline: 150 min maximum

ASSUMED wall-time budget

Elapsed	Task / exit artifact
0–30 min	S2.0: one corrected rerun per failed/marginal S1 branch.
30–65 min	S2.1: E6 two-Pi edge run; selected sweep, not full campaign.
65–95 min	S2.2: E7-C real-UART readiness / bounded trial.
95–130 / 130–150 min	S2.3–6: CN/OS fault demo; verified archive and stop.

If E7/supervisor preparation is missing, document BLOCKED; do not improvise firmware.

S2.0 | Rerun only the earliest failed S1 condition

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Use the first 30 min to revisit the earliest failed Session-1 condition with one specific correction.

Physical setup: Read handoff.md, rebind P0 to the new Session-2 RUN and DATA roots, and recheck power and identity. Renew N1 after a reboot or topology change.

What you see: Each rerun has a fresh ID, the same original acceptance rule and a note naming the one cause changed. Old evidence remains untouched.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: A rerun passes only if it meets the original numeric and observable criteria. Stop retries at minute 30; unresolved work stays OPEN.

If FAIL: Preserve the new failure and defer further WBR work. Continue independent CN tests only if their own power and physical prerequisites are healthy.

Take these files home: Keep DATA/new-retry-ID, the matching RUN logs, fresh receipts and RUN/S2-rerun-ledger.md beside the Session-1 handoff.

S2.0 | Pi / control, fresh bindings (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
export RUN="$HOME/wbr-evidence/study-S2-20261019"
export DATA="$HOME/wbr-takes/study-S2-20261019"
mkdir -p "$RUN" "$DATA"
vcgencmd get_throttled > "$RUN/power-before.txt"
lsusb -t > "$RUN/usb-before.txt"
# Repeat only the failed S1 step, with new output paths/IDs.
# Example ID=C2-ROS-retry01; same S1.5/S1.7 recipe.
# Reboot/topology change: renew N1; never reuse the old receipt.
```

> Use the checks and failure branch on the preceding S2.0 step card.

S2.1 | CN/OS integration: E6 two-Pi edge offloading

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Run E6 on the two real Pis to distinguish client-clock network RTT from server-local queue and service time.

Physical setup: Disconnect the camera. Use the 8 GB Pi B as edge server and the 4 GB Pi A as client, each with its own supply. Connect Ethernet and keep local consoles available.

What you see: Wait for the server's READY line before starting the client. The client writes frames.csv and meta.json; the server writes service-times.csv.

S2.1 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: The ASSUMED smoke check expects 200 terminal outcomes for 20 s at 10 fps, with no unidentified frame. Report actual loss/late replies and require clean power.

If FAIL: Check power, route/interface, server/port, transport and queue/service in order. Retry once with a fresh output; never subtract unsynchronized absolute host times.

Take these files home: Take home client results_hw/E6/study-S2-* and server hw/results/study-S2-* with preflight, model/OS identities, commands and power readings.

S2.1 | 8 GB server B / ~/CN prepared checkout (1/3)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/CN
sudo ip link set eth0 up
sudo ip address replace 192.168.77.2/24 dev eth0
sh hw/preflight.sh eth0 /tmp/e6-server-before.json
python3 hw/edge_server.py --host 192.168.77.2 --port 9000 \
  --transport udp --mode synthetic --sleep-ms 4 \
  --queue latest \
  --service-log hw/results/study-S2-udp/service-times.csv
```

Use the checks and failure branch on the preceding S2.1 step card.

S2.1 | 4 GB client A / local console, after READY (2/3)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/CN
sudo ip link set eth0 up
sudo ip address replace 192.168.77.1/24 dev eth0
ping -c 3 192.168.77.2
python3 hw/sender.py --host 192.168.77.2 --transport udp \
  --frame-bytes 30000 --fps 10 --duration 20 \
  --out-dir results_hw/E6/study-S2-smoke-udp-01
python3 hw/run_e6.py --link ethernet --transport udp \
  --dry-run
```

Use the checks and failure branch on the preceding S2.1 step card.

S2.1 | 4 GB client A / selected E6 pass (3/3)

SOURCE-SUPPORTED procedure / execution OPEN

```
# After reviewed smoke; 30 runs x20s, plus overhead.
python3 hw/run_e6.py --link ethernet --transport udp \
  --output-root results_hw/E6/study-S2-wired-udp-01
# Repeat TCP only after stopping/restarting server as TCP.
# Wi-Fi: existing AP; verify route dev wlan0 before any run.
# Full four-pass /120-run campaign is outside this session.
# Retain both Pi models, supplies, clocks and before/after flags.
```

Use the checks and failure branch on the preceding S2.1 step card.

S2.1W | CN/OS integration: Wi-Fi / ONNX branches

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Keep optional Wi-Fi and ONNX comparisons tied to the transport and compute workload actually used.

Physical setup: Use the existing access point and local consoles. Both endpoints must select the same transport. ONNX requires a prepared compatible model and runtime.

What you see: The route to Pi B says dev wlan0. RTT uses the sender clock; queue and service durations use the server clock. ONNX failures remain visible.

S2.1W | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require the intended route and retained outcomes for every generated frame. An absent model/runtime means ONNX is UNKNOWN; a synthetic run cannot qualify it.

If FAIL: Stop the Wi-Fi branch if the route is wrong; do not reconfigure the AP in lab. If ONNX is unavailable, retain a clearly labeled synthetic result instead.

Take these files home: Use fresh results_hw/E6/study-S2-wifi-* or ONNX run roots. Keep the actual model hash, server service log and both Pi configurations.

S2.1W | 8 GB server B / existing Wi-Fi (1/3)

SOURCE-SUPPORTED procedure / execution OPEN

```
WIFI_B=$(ip -4 -o address show dev wlan0 scope global \
  | awk '{split($4,a,"/"); print a[1]; exit}')
python3 hw/edge_server.py --host "$WIFI_B" \
  --transport udp --mode synthetic --sleep-ms 4 \
  --service-log hw/results/study-S2-wifi/service-times.csv
```

Use the checks and failure branch on the preceding S2.1W step card.

S2.1W | 4 GB client A / use server printed IPv4 (2/3)

SOURCE-SUPPORTED procedure / execution OPEN

```
read -r -p 'Pi B Wi-Fi IPv4 address: ' WIFI_B
ip route get "$WIFI_B"
# Continue only if route is dev wlan0.
python3 hw/sender.py --host "$WIFI_B" --transport udp \
  --frame-bytes 30000 --fps 10 --duration 20 \
  --out-dir results_hw/E6/study-S2-wifi-udp-01
# Repeat as -02/-03 if time; TCP requires matching server.
# Optional ONNX service syntax (separate run/config):
# --mode onnx --model EXISTING_MODEL.onnx --threads 1
```

Use the checks and failure branch on the preceding S2.1W step card.

S2.1W | 8 GB server B / OPTIONAL prepared ONNX branch (3/3)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED installed model/runtime; use a separate run root.
cd ~/CN
test -s offline-bundle/mobilenetv2-12.onnx
sha256sum offline-bundle/mobilenetv2-12.onnx
python3 hw/edge_server.py --host 192.168.77.2 --port 9000 \
  --transport udp --mode onnx \
  --model offline-bundle/mobilenetv2-12.onnx --threads 1 \
  --queue latest \
  --service-log hw/results/study-S2-onnx/service-times.csv
# Client: S2.1 sender, new output name study-S2-onnx-01.
# Do not use unchanged synthetic-4ms E6 sweep config here.
```

> Use the checks and failure branch on the preceding S2.1W step card.

S2.2 | CN/OS integration: E7-C UART gateway trial

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Run E7 option C to add real STM32 UART ingress while measuring UART and network intervals separately.

Physical setup: Use the owner-prepared E7 CubeMX image, gateway Pi A and edge Pi B. Cross the selected 3.3 V UART pins, join ground and confirm the port is not a console.

What you see: CRC-checked image chunks assemble at the gateway and reach the edge. Each unique frame receives a terminal outcome with UART and network fields.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: The design requires unique IDs, one terminal outcome per frame and no corrupt forwarding. Real baud reliability and MCU inference remain UNTESTED until measured.

If FAIL: Check power, ground/pins, baud/console, CRC/chunks and server in order. A missing built image blocks this branch; no flash command may be guessed.

Take these files home: Preserve results_hw/E7/study-S2-edge-udp-01 with args.json, frames.csv and outcomes.jsonl, plus the actual firmware/config hashes.

S2.2 | 4 GB gateway A / ~/CN (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED hardware: owner's E7 firmware and real UART.
python3 e7/gateway/gateway.py --serial /dev/serial0 \
  --baud 115200 --host 192.168.77.2 --port 9000 \
  --transport udp --reply-format timed \
  --outDir results_hw/E7/study-S2-edge-udp-01
# Stop normally with Ctrl-C after the bounded fixture trial.
# Repeat TCP only with matching timed-reply server.
# No --net-status-ms here until a compatible supervisor exists.
```

Use the checks and failure branch on the preceding S2.2 step card.

S2.3 | CN/OS integration: supervisor readiness

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Establish real-UART FRESH before fault injection, so a later state change has a verified healthy starting point.

Physical setup: Use the compatible OS v3 board image, one hea_bridge UART writer and a prepared CN-result IPC producer. Replace the E7 image/path; do not run both UART writers.

What you see: Accepted local HEARTBEAT, INFERENCE and NET_STATUS make MCU_STATUS report FRESH=1. Save the real MCU state and clock telemetry.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Continue only after state 1 is actually observed and recorded. Sources report host/PTY tests only; the real UART path is still UNTESTED.

If FAIL: Stop if the image, producer or FRESH state is missing. E7 telemetry alone sends no INFERENCE and cannot satisfy the supervisor's local readiness checks.

Take these files home: Retain bridge.csv, inf_rx.csv, state_events.csv and readiness notes, together with the exact OS firmware and configuration hashes.

S2.3 | Gateway Pi / prepared OS checkout (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED: deployed image, UART and CN-result IPC producer.
# Source interface exists; this hardware recipe is not validated.
linux/build/hea_bridge --socket /tmp/cn-results.sock \
  --serial /dev/ttyAMA0 --baud 921600 \
  --heartbeat-ms 20 --net-status-ms 20 --policy latest \
  --csv bridge.csv
# Do NOT run E7 gateway as another writer to the same UART.
# CN producer must provide accepted <IQQBBI IPC records.
# No verified producer/flash command is supplied: STOP if absent.
```

Use the checks and failure branch on the preceding S2.3 step card.

S2.4 | CN/OS integration: Wi-Fi loss while heartbeat lives

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Interrupt network results while keeping local heartbeat alive, so the MCU can demonstrate stale-result HOLD followed by latched FAILSAFE.

Physical setup: Start from recorded FRESH on a motor-free bench. Keep local heartbeat at 20 ms and use the transmitting result peer's wlan0, with local console recovery available.

What you see: The source defaults predict HOLD at result/network age $\geq$200 ms and FAILSAFE $\geq$500 ms. These are provisional configuration values, not measured WBR safety bounds.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require recorded state order 1→2→3 with age/silence reasons and no unexplained missing events. A stopped local heartbeat tests a different failure path.

If FAIL: Restore the result path and keep the failed trace. If heartbeat also stopped, label that attempt accordingly; do not change timeouts to obtain the desired sequence.

Take these files home: Keep bridge.csv, state_events.csv, fault-on/off observations and the gateway/server interface and config records. Measure transitions on a supported clock.

S2.4 | Wi-Fi transmitting peer / prepared OS scripts (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED real Wi-Fi/UART integration; source CLI verified.
# Affect outgoing results on edge server B, not management SSH.
scripts/netem_inject.sh --dry-run apply wlan0 --loss 100
# After owner checks the printed direction and local console:
scripts/netem_inject.sh apply wlan0 --loss 100
# Leave >500ms while gateway HEARTBEAT stays alive.
# Observe and save MCU states, then restore the path:
scripts/netem_inject.sh clear wlan0
```

> Use the checks and failure branch on the preceding S2.4 step card.

S2.5 | CN/OS integration: recover and test heartbeat loss

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Demonstrate recovery from latched FAILSAFE, then test heartbeat loss separately so the two mechanisms remain understandable.

Physical setup: Restore the network while FAILSAFE is latched. On the motor-free bench, the owner resets the board; re-establish accepted heartbeat, inference and network status.

What you see: After reset, expect INIT→FRESH. A short HOLD-only interruption may recover automatically; stopping heartbeat instead predicts FAILSAFE near 100 ms.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require FAILSAFE to remain latched on resumed packets alone. Reset and new readiness must precede FRESH. The 100 ms default is provisional, not a guaranteed physical response.

If FAIL: Stop on an unexpected state and save raw protocol/state traces. A missing hardware result leaves robot control independence unproven; do not infer balancing from this bench.

Take these files home: Preserve separate reset/recovery and heartbeat-loss traces, with fault/reset notes and exact thresholds. A bounded short-fault fixture is still UNVERIFIED.

S2.5 | Gateway / manual controlled fault sequence (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
# Restore Wi-Fi result path and preserve prior trace first.
# Owner presses board RESET after latched FAILSAFE.
# Verify fresh HB + INFERENCE + NET_STATUS -> INIT then FRESH.
# For HB-loss case: Ctrl-C the sole hea_bridge process.
# Expected provisional HB timeout: 100ms -> latched FAILSAFE.
# Resume bridge; packets alone do not clear the latch.
# Owner resets board, re-establishes FRESH; save a new trace.
# HOLD-only recovery needs measured <500ms result interruption.
# UNVERIFIED: bounded real-network short fault timing fixture.
```

> Use the checks and failure branch on the preceding S2.5 step card.

S2.6 | CN/OS integration: archive and interpret honestly

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Finish within 150 min with complete node/MCU evidence and an honest account of what was not tested.

Physical setup: Stop jobs normally, clear the injected qdisc, and sample power on each Pi. Keep the Mac reachable while both node archives are copied.

What you see: The cleared qdisc is checked explicitly. The archive contains client outcomes, server service logs, UART states, configs, identities and verified file hashes.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require one recorded outcome per generated frame, valid clock boundaries and matching copies. Missing telemetry stays UNKNOWN and no hardware gate is promoted by this archive.

If FAIL: Keep each node's originals after a failed copy. Retry transfer once; do not delete files until the finalized manifests verify on the Mac.

Take these files home: Take home study-S2-20261019/{wbr,cn-client,cn-server,os} and summary.md, including rerun evidence, fault/recovery traces and unresolved prerequisites.

S2.6 | Each Pi / control (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
vcgencmd get_throttled
vcgencmd pmic_read_adc EXT5V_V
# Save the raw values in that node's run directory.
# On edge peer if injection was used:
scripts/netem_inject.sh status wlan0
# Verify cleared qdisc; do not infer it from restored replies.
# Copy both CN results roots and OS CSVs to Mac; hash finalized files.
# Owner shuts down only after the archive verification.
```

> Use the checks and failure branch on the preceding S2.6 step card.

S2.6 | Mac / both-node CN archive, then hash review (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ARCH="$HOME/Desktop/Research/wbr-data/study-S2-20261019"
mkdir -p "$ARCH/cn-client" "$ARCH/cn-server" "$ARCH/os"
# UNVERIFIED deployment SSH targets/routes: bind from lab inventory.
read -r -p 'Client SSH user@host: ' CN_CLIENT_SSH
read -r -p 'Edge SSH user@host: ' CN_EDGE_SSH
rsync -a --checksum "$CN_CLIENT_SSH:CN/results_hw/" \
  "$ARCH/cn-client/"
rsync -a --checksum "$CN_EDGE_SSH:CN/hw/results/" \
  "$ARCH/cn-server/"
# OS trace dir is a pre-lab binding, not a guessed path.
# Copy bridge/inf_rx/state_events CSVs plus firmware/config hashes.
# Verify per-node finalized SHA256 manifests as in S1.5F.
```

Use the checks and failure branch on the preceding S2.6 step card.

Session 2: what you take home — runs and traces

SOURCE-SUPPORTED inventory / hardware tests conditional

Bag or file family	Home consumer
WBR retry bags, native takes and fresh receipts	Same H0–H6 path as the corresponding S1 take.
Both Pi E6 outcomes/meta/service/preflight logs	H0 archive check; H7 clock/coverage discipline only.
E7 args/frames/outcomes + firmware identity	H0 archive check; separate CN timing analysis.
OS bridge/inf_rx/state events + faults/resets	H0 archive check; separate supervisor transition review.

CN/OS CSVs are not trajectories and must not be fed to H7 ATE/RPE.

SOURCE-SUPPORTED inventory / no new gate evidence

Bag or file family	Home consumer
Power before/after, qdisc status, node configs	H0 preservation; join with separate CN timing review.
Model/binary/firmware hashes and exact commands	H0 integrity; H6 only for an actual WBR replay.
RUN/S2-rerun-ledger, summary and SHA256SUMS	H0 decides usable inputs; H1–H7 skip absent branches.
Independent route/reference, only if acquired	H7 metrics if compatible; otherwise report proxies.

H1–H5 consume sensor data, not E6/E7 logs. Keep full hardware campaigns OPEN.

At home: analyze immutable inputs

SOURCE-SUPPORTED offline workflow

- Begin with copied hashes, bag metadata, raw counts/timestamps and config identities.
- Then validate conversion, fit calibration/noise and replay the same data repeatedly.
- Only independent ground truth permits ATE/RPE; room routes provide proxies.

> Keep acquisition failures visible through every derivative.

H0 | Home: bag metadata, replay and timestamps

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Check recording completeness before analysis; a bag that opens can still be missing data or context.

Physical setup: Use finalized copied bags and sourced Jazzy at home. Stop live camera publishers and replay in the separate localhost domain.

What you see: Bag information lists topics, types and counts. The audit must also examine source headers, clock domains, complete-window gaps and replay contents.

H0 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: The ASSUMED review permits no unexplained loss/reversal and requires complete context plus replay count/header/pixel agreement. Bag-info alone cannot pass it.

If FAIL: Resolve hashes/copy first, then context/QoS and original timestamp metadata. Do not sort, interpolate or fill damaged data to make the report pass.

Take these files home: Keep derived bag-info, timing/count reports and replay logs beside the input hash manifest. Preserve all original bags unchanged.

SOURCE-SUPPORTED procedure / execution OPEN

```
export BAG="$HOME/wbr-takes/study-S1-20261018/S-ROS-01/ros"
ros2 bag info "$BAG"
# Stop live wrapper first; use isolated localhost replay domain.
export ROS_DOMAIN_ID=43 ROS_LOCALHOST_ONLY=1
ros2 bag play "$BAG" --rate 1.0 --delay 3 \
  --qos-profile-overrides-path "$STUDY/lab-config/qos.yaml"
# In another domain-43 terminal: observe source headers/counts.
# UNVERIFIED: extended stereo/metadata bag audit interface.
# n3_bag.py accepts only its reviewed N3 receipt/topic contract.
```

Use the checks and failure branch on the preceding H0 step card.

H1 | Home: native SDK bag audit and EuRoC export

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Identify and audit the native SDK schema before exporting EuRoC input, because a .db3 extension does not define its contents.

Physical setup: On the Mac's canonical WBR checkout, select an actual finalized native take and a fresh derived output root. No camera is needed.

What you see: Inspection and audit produce schema/timing reports. Export creates lossless grayscale PNGs, raw gyro/accel, the combined grid, manifest and times_ns.txt.

H1 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require an allowed common source domain, valid order, no rejected gaps and nonempty image times strictly inside combined IMU coverage.

If FAIL: Stop on unsupported schema, ROS1 magic or invalid timing. Do not rename the input or use a diagnostic override to claim qualification.

Take these files home: Keep DERIVED/C1-native-euroc in full, plus the audit JSON/Markdown. Its manifest links pixels, timestamps and converter identity to the raw hash.

H1 | Mac / canonical WBR checkout (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
export ARCH="$HOME/Desktop/Research/wbr-data/study-S1-20261018"
export NATIVE="$ARCH/native/C1-NATIVE-01/take"
export DERIVED="$HOME/Desktop/Research/wbr-data/derived-study"
python3 tools/take_tools/take_tools.py inspect \
  "$NATIVE/stereo_imu.db3"
python3 tools/take_tools/take_tools.py audit \
  "$NATIVE/stereo_imu.db3" --json /tmp/C1-audit.json \
  --markdown /tmp/C1-audit.md
python3 tools/take_tools/take_tools.py export-euroc \
  "$NATIVE/stereo_imu.db3" "$DERIVED/C1-native-euroc"
```

Use the checks and failure branch on the preceding H1 step card.

H2 | Home: native replica → derived ROS companion

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Test whether a native take can produce a faithful ROS companion through file playback, without two programs opening the camera.

Physical setup: Physically unplug the camera. Prepare the explicit ROS recorder before starting wrapper file playback from the finalized native SDK file.

What you see: The source exposes rosbag_filename, but this target db3's startup and stamp preservation have not been validated. A derived ROS bag is the expected candidate output.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: This route remains UNVERIFIED until complete counts, first/last IDs, original stamps/domains and pixels match exactly. A readable derived bag is insufficient.

If FAIL: Keep the native file and mark conversion BLOCKED after any missing boundary, rewritten stamp or unsupported schema. Do not schedule native replicas until this gap is resolved.

Take these files home: Retain derived/C1-native-ros and a conversion receipt linking its input hash and full-window comparisons. Failed candidate conversions remain evidence.

H2 | Home ROS / file publisher, no live camera (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
# UNVERIFIED: db3 load and lossless full-window behavior.
# Start ros2 bag record with the S1.5 explicit topic list first.
ros2 launch realsense2_camera rs_launch.py \
  config_file:="$STUDY/lab-config/ros-stereo.yaml" \
  rosbag_filename:="$NATIVE/stereo_imu.db3" rosbag_loop:=false
# No tested pause/start handshake exists in this recipe.
# Never claim this derived replay is a second live measurement.
# Native A CSV has no rosbag_filename route: see H3.
```

Use the checks and failure branch on the preceding H2 step card.

H3 | Home: conversion boundary and preparation gaps

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Keep schema boundaries explicit before fitting: native SDK bags, wrapper ROS bags and native IMU CSV are different inputs.

Physical setup: Use the source-supported native S/C converter only on its expected take directory. Have the fixed Foundation path and prepared Kalibr image verified.

What you see: The native helper creates ROS1 take.bag. Ordinary ROS S/C-to-Kalibr and raw A extraction still lack reviewed adapters; the commands mark those gaps UNVERIFIED.

H3 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require matching counts/pixels/SI units and source-stamp error ≤ 1 μs after conversion. An absent extractor blocks that analysis branch.

If FAIL: Stop fitting and prepare the missing conversion at home. Never invent a program or substitute recorder write time for sensor source time.

Take these files home: Keep the native Kalibr output, input/output hashes and count/stamp/pixel receipt. Preserve every rejected derivative separately.

H3 | Mac / native S/C only (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/Desktop/Research/wbr-perception-phase1-runtime
bash calibration/lab/mac/take_to_kalibr_bag.sh \
  "$NATIVE" "$DERIVED/C1-native-kalibr"
# Repeat independently for C2, C3 and S native takes.
# UNVERIFIED: wrapper ROS bag -> Kalibr source-stamp exporter.
# UNVERIFIED: ROS raw IMU -> Allan CSV extraction.
# UNVERIFIED: native Allan / native N2 -> ROS companion.
# No executable for those absent interfaces is claimed.
```

Use the checks and failure branch on the preceding H3 step card.

H4 | Home: Kalibr C fits and stereo validation

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Compare residuals and independent repeats before selecting a spatial/time candidate for later review.

Physical setup: Use audited native-to-ROS1 bags, the physically measured target YAML and the prepared Intel-Mac Kalibr image. Keep factory operating settings unchanged.

What you see: Each C fit produces a report, results and camera chain. The S stereo check produces its own report; ORB YAML files remain candidates.

H4 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: ASSUMED review: radial RMS $\leq$0.5 px, C1–C3 spread $\leq$3 mm/1° and time-shift spread $\leq$1 ms. S validation baseline should differ by $\leq$1 mm.

If FAIL: Inspect timing, target/light, motion, transform direction and noise in order. Do not replace factory settings based on an attractive single fit.

Take these files home: Keep DERIVED/C1-imucam through C3 and S-stereo with PDFs, logs and candidate files. Record noise source and every failed detection/residual pattern.

H4 | Mac / canonical WBR checkout (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
export TARGET="$HOME/Desktop/Research/wbr-data/measured-target.yaml"
bash calibration/lab/mac/run_kalibr_camimu.sh \
  "$DERIVED/C1-native-kalibr/take.bag" "$TARGET" \
  "$DERIVED/C1-imucam"
# Repeat with C2/C3 input/output names, no overwrites.
# Optional fourth arg is a supported measured IMU noise YAML.
bash calibration/lab/mac/run_kalibr_stereo_check.sh \
  "$DERIVED/S-native-kalibr/take.bag" "$TARGET" \
  "$DERIVED/S-stereo"
```

> Use the checks and failure branch on the preceding H4 step card.

H5 | Home: Allan fitting on native raw channels

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Fit only the noise processes supported by long stationary raw data; unsupported walks must remain UNKNOWN.

Physical setup: Use complete native raw A CSV and its power metadata. Primary ROS A first needs the missing reviewed H3 extraction; do not pass a ROS bag directory to the native analyzer.

What you see: The analyzer emits per-axis Allan plots, fit ranges/counts and raw/x10 YAML. Null unsupported coefficients are unknown values, not zeros.

H5 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: ASSUMED fits need slopes −0.5/+0.5 within ±0.1 for ≥1 decade, $R^2 \geq 0.95$ and ≥30 clusters per point. Compare halves; estimates should agree within 30%.

If FAIL: Keep unsupported walk UNKNOWN. Split invalid timing segments without resampling across gaps, and replace power-damaged data with a fresh take.

Take these files home: Keep allan.json, summary.md, imu_allan_raw.yaml, imu_allan_x10.yaml and plots under the derived analysis root, linked to raw-data hashes.

H5 | Mac / native A only after preservation requirement resolved (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 calibration/allan/allan_analyze.py \
  "$HOME/Desktop/Research/wbr-data/allan-A-native-15h-01" \
  "$DERIVED/A-native-15h-analysis"
# Do not pass an ordinary ROS bag directory to this command.
# Null YAML scalars are UNKNOWN and cannot feed Kalibr/ORB.
# Review raw native accel100 / gyro200, not interpolated accel.
# First/second-half stability comparison is not automated here.
```

> Use the checks and failure branch on the preceding H5 step card.

H6 | Home: offline ORB before any live experiment

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Replay one immutable motion take before live trials, so build, input and estimator failures can be separated.

Physical setup: Use a prepared home Pi, audited EuRoC export and reviewed settings. Source rules still require explicit ORB authorization. The optional demo branch needs a real existing hand100 take.

What you see: A successful trial writes pose/state evidence and logs; a failed trial retains its crash or empty output. A still calibration take is not a qualified route test.

H6 | Decision, recovery and files

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: ASSUMED route review proposes initialization $\leq$30 s, one map, zero resets, loss $\leq$1% and no lost interval >1 s. Missing route data or events leaves this review OPEN.

If FAIL: Inspect hashes/export, build identity, coverage, calibration and motion in order. Keep failures; do not tune around an input defect or rename a take to fit the demo wrapper.

Take these files home: Keep exact commands, input/settings/binary hashes, trajectories, state/power logs and demo plots/stats when applicable. Fetch raw native bags separately.

H6 | Mac / optional existing demo take, not a C/S path (1/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
cd ~/Desktop/Research/wbr-perception-phase1-runtime
export WBR_HOST=richard@192.168.50.2
# Requires an actual completed native demo record hand100.
# Do not rename a calibration take to satisfy this wrapper.
bash experiments/2026-10-15-demo-day/demo.sh offline hand100
bash experiments/2026-10-15-demo-day/demo.sh fetch hand100
bash experiments/2026-10-15-demo-day/demo.sh report hand100
```

> Use the checks and failure branch on the preceding H6 step card.

H6 | Home Pi / generic exported immutable take (2/2)

SOURCE-SUPPORTED procedure / execution OPEN

```
# Prepared built engine/settings; owner ORB authorization required.
export ORB_DIR="$HOME/orbslam3/ORB_SLAM3"
export TAKE_DIR="$HOME/wbr-data/derived/V1-native-euroc"
export SETTINGS_FILE="$HOME/wbr-data/reviewed-settings.yaml"
export LD_LIBRARY_PATH="$ORB_DIR/lib:$ORB_DIR/Thirdparty/DBoW2/lib"
export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:$ORB_DIR/Thirdparty/g2o/lib"
"$ORB_DIR/Examples/Stereo-Inertial/stereo_inertial_euroc" \
  "$ORB_DIR/Vocabulary/ORBvoc.txt" "$SETTINGS_FILE" \
  "$TAKE_DIR" "$TAKE_DIR/times_ns.txt"
# Run in a fresh directory; archive exact binary/library hashes.
# Required V route take may still be absent: keep replay OPEN.
```

Use the checks and failure branch on the preceding H6 step card.

H7 | Home: trajectory evaluation only with a reference

SOURCE-SUPPORTED procedure / execution OPEN

Goal / why: Evaluate trajectories only after confirming that estimate and independent reference describe the same physical motion and compatible epochs.

Physical setup: Provide actual paths and verified frame/origin/clock declarations. For MH01 use its dataset body labels; the displayed imu_camera template is for a compatible camera reference.

What you see: The evaluator writes association and declared rigid-alignment ATE/RPE results. It must report coverage, exclusions and failures alongside plots.

SOURCE-SUPPORTED procedure / execution OPEN

PASS / FAIL: Require finite associated data and supported frame/clock meaning. Without independent ground truth, skip ATE/RPE and report closure or measured-segment proxies instead.

If FAIL: Inspect units/order, pose direction/extrinsics, epochs, association and alignment in order. Do not fit scale to hide a metric-stereo error.

Take these files home: Preserve DERIVED/traj-eval with raw inputs/hashes, exact arguments, assumptions and all excluded-pair counts.

H7 | Mac / replace only verified input paths (1/1)

SOURCE-SUPPORTED procedure / execution OPEN

```
python3 tools/trajectory_eval/traj_eval.py single \
  --est "$DERIVED/CameraTrajectory.txt" \
  --ref "$HOME/Desktop/Research/wbr-data/reference.csv" \
  --est-format orbslam3 --ref-format euroc_gt \
  --clock-assumption 'Verified common sensor epoch; zero offset' \
  --est-frame 'imu_camera' --ref-frame 'imu_camera' \
  --t-max-diff 0.02 --align se3 \
  --rpe-delta 1 --rpe-unit s --out "$DERIVED/traj-eval"
# Input declarations above must be established, not assumed true.
# No independent room reference: skip, report closure/scale proxies.
```

> Use the checks and failure branch on the preceding H7 step card.

Home ablation: preserve every controlled variable

ASSUMED calibration-plan protocol

- V1–V3 are separate textured-room 90 s route takes, not target-only C takes.
- Pair factory vs selected spatial/time settings with raw vs x10 noise on identical inputs.
- Three replays per cell; record initialization, maps/resets, tracking loss and route proxies.

> Select on development takes, verify on held-out V; keep ties and failures.

Backup: printable Session 1 checklist

ASSUMED printable one-page field checklist

- ☐ P0/S1.0–2: ready paths; manifest; flags 0x0; camera branch ≥5000M; serial.
- ☐ S1.3–4: 20/105/300 s + actual review; N3 record/replay or explicit blocker.
- ☐ S1.5–8: ROS S/C1/C2/C3 + metadata; A pilot; native branches only if companion ready.
- ☐ S1.9–10: camera off, correct bench firmware, link results; hashes/handoff; 150 min.

Stop on power/USB/order faults. One new-ID retry; no gate PASS without evidence.

Backup: printable Session 2 checklist

ASSUMED printable one-page field checklist

- ☐ S2.0: handoff read, fresh power/identity/N1 as needed; retries stop at minute 30.
- ☐ S2.1: 8 GB server / 4 GB client; E6 exact service/transport, outcomes and both logs.
- ☐ S2.2–5: E7 image ready; real UART FRESH; result loss vs HB loss; latch/reset contract.
- ☐ S2.6: qdisc cleared, both-node/MCU traces hashed, unresolved work OPEN; 150 min.

No FRESH or no verified firmware/producer means skip the fault demo and retain BLOCKED.

Backup: what the two-session plan cannot promise

OPEN / UNVERIFIED

Gap	Required before acceptance
N3 card receipt policy	Reviewed runner/receipt compatible with approved 128 GB exception.
Native / ROS companions	Validated source-preserving playback; CSV adapters absent.
Primary ROS → analysis	Reviewed wrapper-bag→Kalibr / raw Allan extraction.
Real CN/OS demo	Built v3 board image, result IPC producer and UART qualification.

These are source/tooling gaps, not commands to improvise at the bench.

Sources and reading order

SOURCE-SUPPORTED provenance

Local source	Use
base/main-v4-91p.tex → base/main.tex	Lecture flow → Phase-8 factual precedence.
sources/LAB_DAY…4b63.md + STM32_LINK…	Latest lab scripts, native diagnostics, link bench.
docs/D435I_CALIBRATION_PLAN.md	Current storage/target/timing/noise review protocol.
CN hw/e7 + OS-wt-cn CN_INTEGRATION	E6/E7 interfaces; v3 supervisor behavior.

Full paths, hashes and material conflicts are in qa/SOURCES.md and README.

Review standard: explain the complete evidence chain

KNOWN method

- What is known, measured, assumed and still OPEN?
- Which frame and clock does each quantity belong to?
- What exact observation rejects this take or triggers fallback?
- Where are the immutable raw data, hashes, config and failure logs?

> A complete build can pass document QA while every hardware gate remains OPEN.